

AE 4311 Aircraft Design

Final Report

Conceptual Design of a
Middle-of-the-Market Commercial Airliner

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“I certify that I have abided by the honor code of the Georgia Institute of Technology and followed the collaboration guidelines as specified in the project description for this assignment.”

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1 Mission and Requirements

1.1 Project Overview

This report documents the conceptual design of a middle-of-the-market commercial airliner. The design process primarily follows the methodology developed by Roskam [1] and supplemented by Nicolai and Carichner [2]. The process started with a structured set of requirements derived from the mission specification and progressing through configuration selection, weight and constraint sizing, cost estimation, airfoil and wing analysis, fuselage and tail sizing, and an initial weight and balance analysis. Each stage of the process informed the next, and necessary iterations were made when later analysis resulted in different inputs. The report reflects the final converged design state of the aircraft.

1.2 Design Mission Specification

The design mission calls for a two turbofan powered commercial airliner certified under FAR Part 25, capable of carrying 280 passengers in a single-class arrangement over a design range of 3,500 nm at a cruise Mach number of 0.80 and 35,000 ft. Maximum operating speed is Mach 0.82, and service ceiling is 40,000 ft. The aircraft must take off in less than 9,000 ft over a 35 ft obstacle from a sea-level runway, approach at no more than 145 kt, sustain a 30° bank turn at 250 kt and 5,000 ft, and maintain a 2.4% climb gradient with one engine inoperative (OEI). Reserve requirements consist of 30 minutes of loiter plus a 200 nm diversion at 240 kt and 5,000 ft. The flight crew comprises two pilots and four cabin crew, and the crew and passengers are each sized at 175 lb of body weight plus 30 lb of baggage. The full restructured set of requirements derived from this specification is presented in Section 1.4.

1.3 Similar Aircraft

A survey of in-service and recently retired aircraft with a similar mission profile was conducted to inform later sizing assumptions and design choices. Twelve aircraft were selected, spanning a maximum takeoff weight (MTOW) range of roughly 100,000 lb to 550,000 lb. The compiled data are summarized in Table 1, with values drawn primarily from specifications in Jane's *All the World's Aircraft* [3, 4].

Table 1. Comparable middle-of-the-market commercial aircraft.

Aircraft	MTOW (lb)	OEW (lb)	Pax (max)	Range (nm)	TOFL (ft)	Span (ft)	AR	Thrust (lbf)
Boeing 767-200ER	395,000	185,900	290	6,545	8,150	156.1	7.99	2×57,900
Airbus A330-200	524,700	265,655	293	7,200	8,700	197.8	10.06	2×72,000
Ilyushin Il-96-300	551,000	265,198	330	5,400	8,858	197.8	9.50	4×35,300
Tupolev Tu-214	242,508	129,852	210	3,500	8,200	138.0	9.00	2×35,300
Boeing 787-8	502,500	261,100	296	8,000	8,500	197.3	11.00	2×64,000
Irkut MC-21-300	174,700	93,000	211	3,500	6,900	116.0	11.50	2×31,000
Airbus A310-300	361,550	194,000	240	4,000	8,000	144.0	8.80	2×59,000
Boeing 777-200	545,000	298,000	440	5,200	8,900	199.9	8.68	2×77,000
Boeing 737-900ER	187,700	98,000	220	3,200	7,500	117.5	9.45	2×27,300
COMAC C919	160,000	89,000	174	2,500	6,600	117.5	9.50	2×30,000
Tupolev Tu-334	105,600	65,000	126	1,900	6,200	95.6	8.93	2×16,550
Airbus A321-200	206,100	110,000	240	3,200	7,000	117.5	9.45	2×33,000

It can be observed from the list of similar aircraft that twin-turbofan layouts are most common. MTOW scales roughly linearly with passenger count and design range, and operating empty weight (OEW) accounts for around 50–55% of MTOW. Aspect ratios are within the range of 8–11, with newer aircraft such as the 787-8 and MC-21 reaching the upper end. Takeoff field lengths between 6,600 ft and 8,900 ft meet the design requirement of 9,000 ft. The 767-200ER and A310-300 are the closest matches to the design point in terms of passenger capacity and MTOW class, and their aspect ratios (7.99 and 8.80, respectively) and wing loadings (~ 130 psf at MTOW) are used as reference points for early sizing assumptions in later sections.

1.4 Hierarchical Requirements

The mission specification was restructured into a hierarchical requirements tree under a single Level 1 capability requirement. The Level 1 requirement is defined by what the aircraft is designed to carry and the main point where it will fly. The remaining constraints are treated as supporting requirements and are organized in the hierarchy below.

- **Level 1 – Primary Capability:** The aircraft shall transport 280 passengers over a design range of 3,500 nm at a cruise Mach number of 0.80, powered by two turbofan engines.
 - **Level 2 – Payload:** The aircraft shall accommodate 280 passengers in a single-class arrangement.
 - * **Level 3:** Per-passenger payload shall be 205 lb, consisting of a 175 lb occupant and 30 lb of baggage.

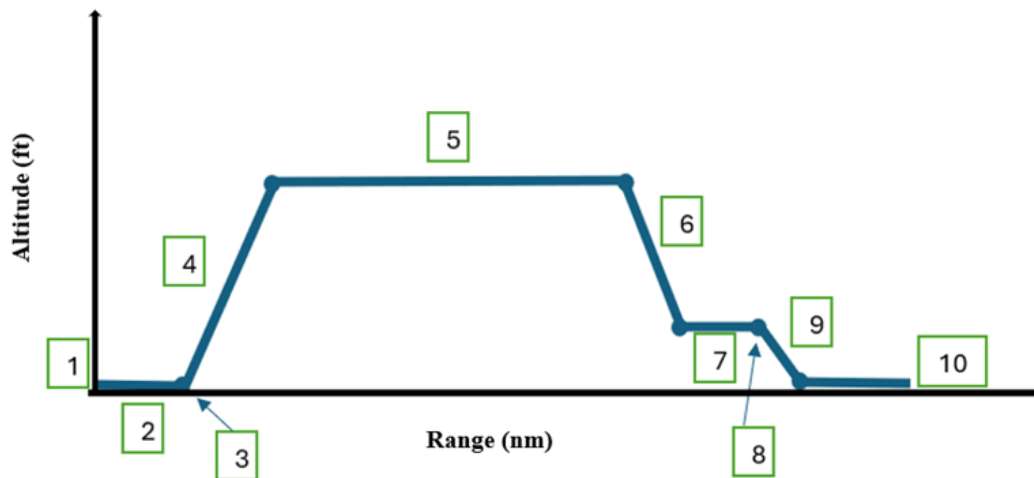
- * **Level 3:** The cargo hold shall provide sufficient volume to accommodate 8,400 lb of passenger baggage.
- **Level 2 – Range and Cruise:** The aircraft shall achieve a design range of 3,500 nm at a cruise altitude of 35,000 ft and a cruise Mach number of 0.80, with a maximum operating Mach number of 0.82 and a service ceiling of 40,000 ft.
 - * **Level 3:** The cabin pressurization system shall maintain an effective cabin altitude of 8,000 ft when operating at the service ceiling.
- **Level 2 – Takeoff and Landing:** The aircraft shall be compatible with standard sea-level airport infrastructure.
 - * **Level 3:** Takeoff field length shall not exceed 9,000 ft over a 35 ft obstacle at sea level.
 - * **Level 3:** The aircraft shall maintain a climb gradient of at least 2.4% with one engine inoperative, landing gear retracted, and flaps extended.
 - * **Level 3:** Approach speed shall not exceed 145 kt at maximum landing weight.
 - * **Level 3:** The aircraft shall sustain a 30° bank turn at 250 kt and 5,000 ft.
- **Level 2 – Crew:** The cockpit and cabin shall accommodate the required flight and cabin crew and provide adequate emergency egress.
 - * **Level 3:** Crew complement shall consist of two pilots and four cabin crew, each sized at 205 lb including baggage.
 - * **Level 3:** The aircraft shall be equipped with sufficient Type A or Type I exits to evacuate all 286 occupants within FAR Part 25 evacuation time limits.
- **Level 2 – Reserves:** The fuel system shall carry reserves sufficient for a 200 nm diversion at 240 kt and 5,000 ft, followed by 30 minutes of loiter at the diversion airport.
- **Level 2 – Certification:** The aircraft shall be designed and certified in accordance with FAR Part 25 standards.

1.5 Design Mission Profile

The design mission profile derived from the specification is summarized in Table 2 and is shown graphically in Figure 1. The profile consists of 10 segments: engine start, taxi out, takeoff, climb, main cruise, descent, reserve cruise, reserve loiter, descent, and landing/taxi in. The mission profile defined here forms the basis for the segment fuel fraction analysis used in the later weight sizing calculation.

Table 2. Design mission profile segments.

Number	Segment	Altitude (ft)	Speed / Condition
1	Engine Start	0	0
2	Taxi Out	0	~15 kt
3	Takeoff	0	Max climb thrust
4	Climb	0 $\rightarrow$ 35,000	To cruise altitude
5	Main Cruise	35,000	Mach 0.80; 3,500 nm
6	Descent	35,000 $\rightarrow$ 5,000	Idle
7	Reserve Cruise	5,000	240 kt; 200 nm
8	Reserve Loiter	5,000	30 minutes
9	Descent	5,000 $\rightarrow$ 0	Approach speed
10	Landing/Taxi In	0	~15 kt

**Figure 1.** Design mission profile map.

1.6 Requirements Conclusion

The mission specification and hierarchical set of requirements established in this section provide the design space for the middle-of-the-market commercial airliner. The requirements were organized under the primary capability of 280 passengers over 3,500 nm at Mach 0.80 and will drive the subsequent sizing decisions. The survey of similar aircraft showed that most aircraft of this class have low-wing configurations with aspect ratios in the range of 8–11 and wing loadings near 130 psf. This provides a baseline that is used throughout the design process.

2 Configuration Selection

2.1 Configuration Introduction

The primary goal of this section is to compare different aircraft configurations that align with a middle-of-the-market commercial airliner mission specification. Figures of merit (FOM) are established to compare the configuration alternatives based mostly on qualitative assessments tied to the requirements. From there, the designer can assign weights to the FOMs according to requirements/stakeholders' importance. A weighted Pugh process can be used to identify a “leading” alternative in terms of its scores for each FOM.

2.2 Figures of Merit Selection

In the process of selecting the FOMs, it is necessary first to identify the key stakeholders. The three main stakeholders that were considered for the commercial airliner design were: the airline, the passenger, and the manufacturer. The airline considerations included efficiency (efficiency is a big driver for operating cost), maintenance, and pilot operations. Considerations for the passenger included comfort and quality. Finally, manufacturer considerations included airframe complexity and production cost/time. In addition to the stakeholders, the FOMs were selected to align with commercial airliner requirements while not presenting constraints, to allow for variety within the alternatives' scoring.

This criterion provided the following set of relevant Figures of Merit:

- Aerodynamic Efficiency (weight 0.30)
- Structural Weight (weight 0.20)
- Noise / Ride Quality (weight 0.15)
- Manufacturing Ease (weight 0.20)
- Controls Complexity (weight 0.15)

The aerodynamic efficiency of the aircraft, measured by L/D , drives the fuel burn, which is tied to the requirements for range and OEI climb gradient. Structural weight is selected as an FOM, as light structures could equal more payload, less fuel, or less cost as fewer parts are needed. This ties into the requirements regarding payload as well as takeoff field length. Noise/ride quality is an important metric as the commercial mission specification targets up to 280 passengers. Commercial passengers demand comfort, and this can be a major selling point for the aircraft. Manufacturing ease takes into account the airframe and its complex curves and/or difficult joints. Manufacturing stakeholders aim to minimize production cost and time, and airlines want a lower purchase price. Finally, controls complexity measures the stability of the airframe and the intricacy of the flight control system. This is tied to the requirements made in FAR Part 25 as well as the pilot's training.

2.3 Configuration Alternatives

Three possible configuration alternatives were proposed using an Analysis of Alternatives shown in Table 3. As the middle-of-the-market commercial aircraft prioritizes safe and comfortable travel for many passengers, the design alternatives do not feature any extreme deviations in configuration from the typical in-service commercial aircraft.

Table 3. Commercial airliner configuration alternatives comparison.

Category	Alternative 1	Alternative 2	Alternative 3
Wing attachment	Low wing	Low wing	High wing
Wing add-on	Winglets	None	Struts
Tail type	Conventional	V-tail	T-tail
Engine location	Under-wing	Rear fuselage	Under-wing
Landing gear	Tricycle, wing-mounted	Tricycle, wing-mounted	Tricycle, fuselage-mounted

Alternative 1 was selected as a representation of the current industry standard for middle-of-the-market commercial aircraft. The configuration focuses on low risk and structural and control efficiency. The low-wing attachment paired with the under-wing engine location theoretically maximizes structural efficiency as the engines' weight counteracts the lift force, providing bending relief. This could also allow for a lighter wing structure. The conventional tail allows for manufacturing simplicity and a less complex control design. The winglets can make up for some aerodynamic performance by reducing induced drag from minimizing wingtip vortices. Finally, the low-wing allows for a wing-mounted landing gear, meaning less complex structures are needed to extend the wheels of the aircraft (also meaning less drag). A sketch created using OpenVSP to visualize Alternative 1 is shown in Figure 2.

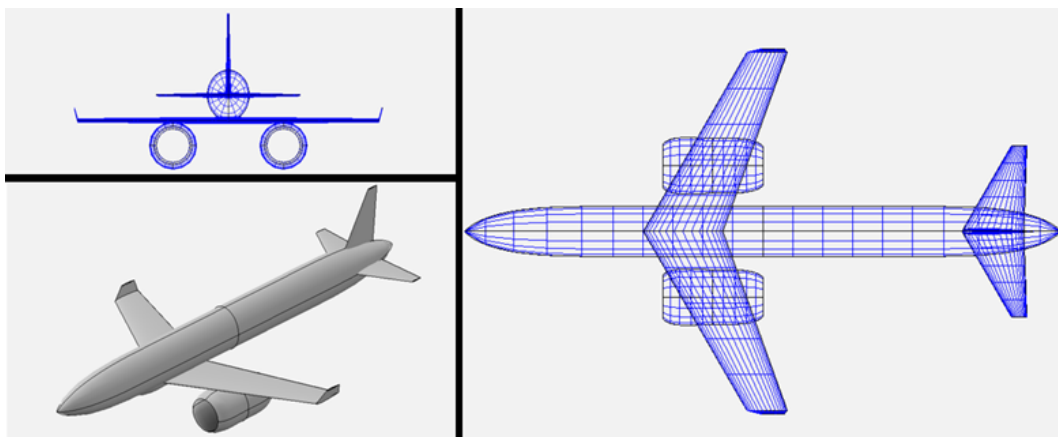


Figure 2. Alternative 1 visualization.

Alternative 2 was selected to provide a more premium passenger experience and maximize

aerodynamic flow. Similar to Alternative 1, a low-wing attachment was selected, which avoids wing structures cutting into the cabin. The low-wing attachment again allows for a simpler wing-mounted landing gear. However, for Alternative 2, the engine location is relocated to the rear fuselage. This promotes smoother laminar flow over the wing. Additionally, cabin noise is significantly reduced as a result. Finally, a V-tail was selected to complement the rear engines, as it provides clearance for the engine exhaust. Also, the V-tail aims to potentially offset the weight penalty of moving the engine location to the rear fuselage. A sketch created using OpenVSP to visualize Alternative 2 is shown in Figure 3.

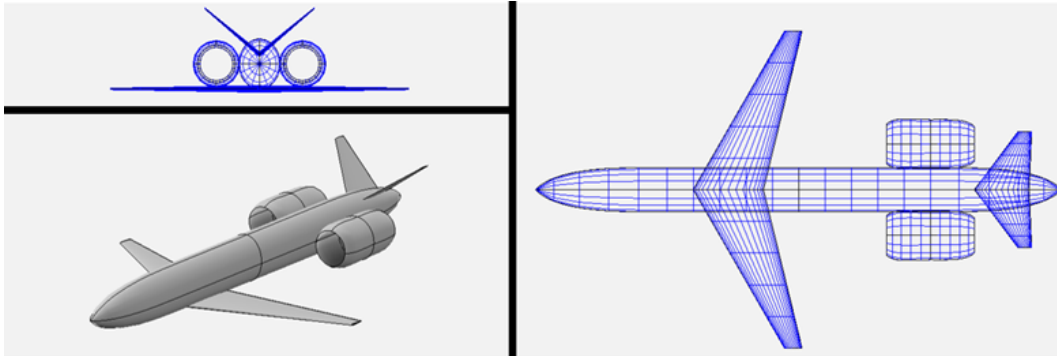


Figure 3. Alternative 2 visualization.

Alternative 3 steers away from industry-standard commercial airliners, but aims to maximize fuel efficiency. A high-wing attachment supported by struts allows for very high aspect ratio wings while preventing wing flutter. The high-wing attachment also allows for the under-wing turbofan engines to be of larger diameter. These larger-diameter engines could mean higher bypass ratios and an increase in propulsion efficiency. A T-tail is used in this alternative to ensure the horizontal stabilizer is placed clear of the downwash generated from the high wing. Finally, due to the height of the wings, the landing gear must be mounted to the fuselage externally, which adds drag. A sketch created using OpenVSP to visualize Alternative 3 is shown in Figure 4.

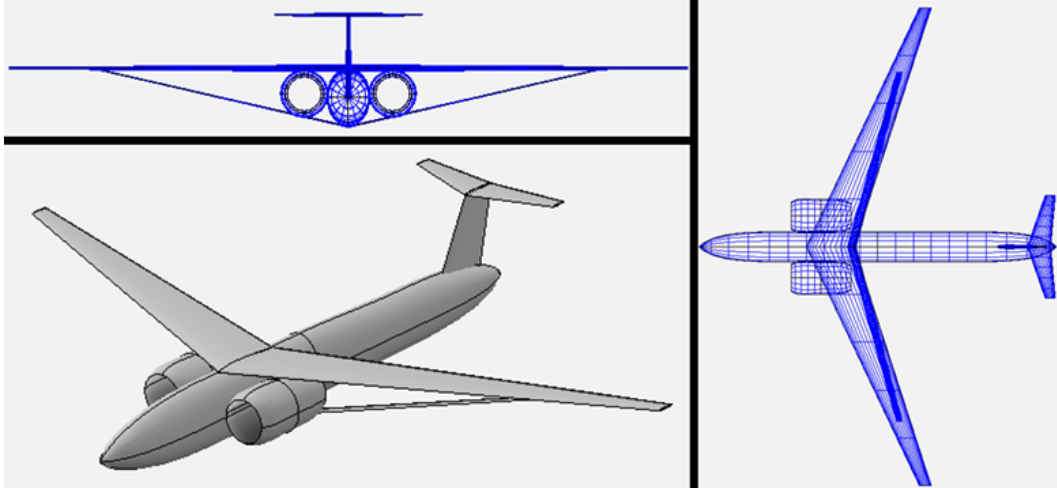


Figure 4. Alternative 3 visualization.

2.4 Configuration Down-Selection

Taking into account the benefits and disadvantages of each, the alternatives are scored against the set FOMs in a weighted Pugh matrix shown in Table 4. The weights of the FOMs were determined based on the main priority of efficiency and passenger ride quality. Structural weight and manufacturing considerations came next, followed by the controls complexity.

Table 4. Weighted Pugh matrix.

Figure of Merit	Weight	Alt. 1	Alt. 2	Alt. 3
Aerodynamic Efficiency	5	3	4	5
Structural Weight	3	3	2	1
Noise / Ride Quality	4	3	5	2
Manufacturing Ease	3	3	2	1
Controls Complexity	2	3	2	2
Total	17	51	56	43

The scoring rationale for each FOM is as follows. For aerodynamic efficiency, Alternative 3 scores highest due to the high-aspect-ratio strut-braced wing, which minimizes induced drag; Alternative 2 scores second as the rear-mounted engines allow for a clean wing surface; Alternative 1 scores lowest as winglets only partially make up for the induced drag penalty of a conventional under-wing layout. For structural weight, Alternative 1 scores highest because the under-wing engines provide bending relief to the wing; Alternative 2 scores second due to the structural penalty from the aft-mounted engines shifting the center of gravity rearward; Alternative 3 scores lowest due to the added weight of struts and fuselage-mounted landing gear. For noise and ride quality, Alternative 2

scores highest as rear-mounted engines significantly reduce cabin noise; Alternative 1 scores second as under-wing engines produce moderate cabin noise; Alternative 3 scores lowest as the high-wing strut configuration could create structural vibrations, on top of having under-wing engines. For manufacturing ease, Alternative 1 scores highest due to its conventional layout and established manufacturing practices; Alternative 2 scores second due to the complexity of a non-standard tail and aft engine arrangement; Alternative 3 scores lowest because the strut, T-tail, and fuselage-mounted gear add significant manufacturing complexity. For controls complexity, Alternatives 2 and 3 score equally as both require more complex control systems than the conventional layout of Alternative 1.

2.5 Configuration Conclusion

Alternative 2 (consisting of a low-wing, V-tail, rear-fuselage-mounted twin-turbofan configuration with wing-mounted tricycle landing gear) scored the highest total score from the weighted Pugh matrix and was selected as the final configuration. This configuration prioritizes the primary stakeholder priorities of aerodynamic efficiency and passenger comfort. The rear fuselage engine location provides a clean wing surface and reduces the noise experienced by the cabin. Despite a structural weight penalty and added manufacturing complexity, Alternative 2 is selected as these factors were weighted less than the aerodynamic efficiency and passenger marketability. This configuration serves as the basis for subsequent sizing and design analyses presented in this report.

3 Weight Sizing

3.1 Weight Sizing Introduction

The primary goal of this section is to determine a maximum takeoff weight of a middle-of-the-market commercial airliner based on methods in Roskam Part I [1]. This method includes performing a weight regression using a compiled list of maximum takeoff weight and empty weight for similar mission profile aircraft. The A and B coefficients obtained from the weight regression, along with assumptions made to calculate mission fuel fractions for climb, cruise, and other segments, are used to calculate a temporary value for the empty weight. This temporary value is converged with the allowable empty weight calculated from the weight regression coefficients to produce a converged maximum takeoff weight. The guessed takeoff weight is also used to construct a Class 1 drag polar, which is used to calculate the cruise L/D ratio. Finally, the calculated L/D value is converged with the previously assumed value to iteratively recalculate the final maximum takeoff weight.

3.2 Weight Regression

The weight regression is built using the maximum takeoff weight and empty weight of several similar aircraft to the commercial airliner mission specification. This data is used to obtain coefficients A and B (by linearizing the $\log_{10}$ of maximum takeoff weight and empty weight values) which define a

realistic empty weight for a given takeoff weight. The weight data of the similar aircraft used in the regression are the same as those listed in Table 1 in Section 1.3.

Plotting the maximum takeoff and empty weights and identifying the linear logarithmic trend resulted in the weight regression plot shown in Figure 5. The calculated A and B values are shown in Table 5.

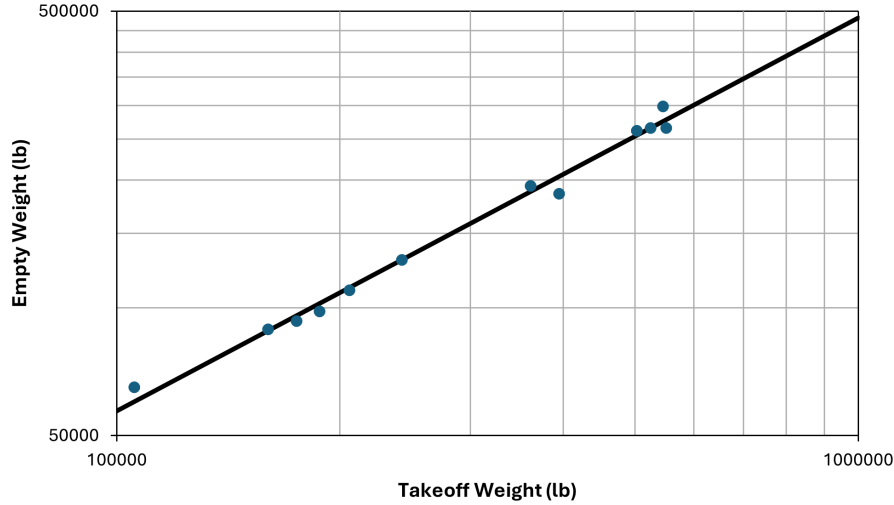


Figure 5. Weight regression of similar aircraft, $\log_{10}$ scale.

Table 5. Weight regression A and B coefficients.

A	B
-0.1362	1.0797

3.3 Mission Profile Assumptions

In determining the calculated empty weight, the crew weight, payload weight, and fuel weight (which includes trapped fuel and oil weight) are subtracted from the guessed takeoff weight value. The crew weight and payload weight are defined in the mission specification; no assumptions are to be made. The fuel weight is estimated by determining mission fuel fractions for each segment along the mission profile of the aircraft. Each individual segment fuel fraction is then combined to yield a total mission fuel fraction which gives the end mission weight divided by the start weight. The mission profile is split into 10 segments and is described in Table 6.

Table 6. Mission profile used for weight sizing.

Number	Segment	Altitude (ft)	Speed / Condition
1	Engine Start	0	0
2	Taxi Out	0	~15 kt
3	Takeoff	0	Max climb thrust
4	Climb	0 $\rightarrow$ 35,000	To cruise altitude
5	Main Cruise	35,000	Mach 0.80; 3,500 nm
6	Descent	35,000 $\rightarrow$ 5,000	Idle
7	Reserve Cruise	5,000	240 kt; 200 nm
8	Reserve Loiter	5,000	30 minutes
9	Descent	5,000 $\rightarrow$ 0	Approach speed
10	Landing/Taxi In	0	~15 kt

The mission fuel fractions for the non-cruise/climb/reserve segments (Engine Start, Taxi, Takeoff, Descent, and Landing/Taxi) are taken from Roskam’s table of suggested fuel fractions for the airplane type of Transport Jets. The fuel fractions for the climb, main cruise, reserve cruise, and reserve loiter are approximated further.

The Breguet Range Approximation is used to calculate the fuel fractions for climb and cruise. For the main cruise segment, the inputs required included: cruise range, cruise velocity, cruise L/D , and cruise TSFC. From there, the cruise mission fuel fraction is calculated. The input and result values of calculating the main cruise mission fuel fraction are shown in Table 7. Cruise range and velocity were directly specified by the mission profile and requirement. The cruise range was given as 3,500 nm, and the velocity was found by converting Mach 0.80 to knots at the cruise altitude of 35,000 ft. The cruise L/D was calculated and converged via a Class 1 drag polar, which is described in a further section. Finally, the cruise TSFC was estimated to be similar to the GE CF6-80C2 turbofan engine cruise TSFC of 0.576 [5], which was used as the powerplant for several similar-class aircraft such as the Boeing 767 and Airbus A310.

Table 7. Cruise mission fuel fraction inputs and result.

Inputs				Output
Range (nm)	Velocity (kt)	L/D	TSFC (lb/lbf/hr)	Cruise FF
3,500	465.4	17.36	0.55	0.791

For the climb segment, the inputs required included: climb altitude, climb rate, climb L/D , and climb TSFC. From there, endurance is determined, and finally, the climb mission fuel fraction is

calculated. The input and result values of calculating the climb mission fuel fraction are shown in Table 8. Climb altitude was directly specified by the mission profile and requirements; however, the remaining inputs required assumptions to be made. The climb rate was estimated as 2,500 ft/min based on the rate of climb of similar aircraft. The climb L/D was estimated by taking 70% of the converged cruise L/D from the drag polar calculations, due to the higher angle of attack during climb. Finally, the climb TSFC was again estimated using the cruise TSFC of the GE CF6-80C2, though choosing a slightly higher value to account for less fuel efficiency during climb.

Table 8. Climb mission fuel fraction inputs and results.

Inputs				Outputs	
Altitude (ft)	Climb rate (ft/min)	L/D	TSFC (lb/lbf/hr)	E (hr)	Climb FF
35,000	2,500	13	0.6	0.233	0.989

The reserve cruise segment was modeled similarly to the main cruise segment described above. The cruise range and cruise velocity were changed to the specified values in the mission profile (200 nm and 240 kt). Again, the cruise TSFC of the GE CF6-80C2 was considered, though choosing a slightly higher value for the reserve cruise TSFC given that this mission segment would be performed at significantly lower altitude. Finally, the reserve cruise L/D was calculated and converged through a separate Class 1 drag polar for the varied reserve cruise conditions (described in a further section). The input and result values of calculating the reserve cruise mission fuel fraction are shown in Table 9.

Table 9. Reserve cruise mission fuel fraction inputs and result.

Inputs				Output
Range (nm)	Velocity (kt)	L/D	TSFC (lb/lbf/hr)	Reserve cruise FF
200	240	18.20	0.6	0.973

The reserve loiter segment was modeled as a simplified version of the climb segment, as the endurance E is a known requirement of 0.5 hours. The reserve loiter TSFC was selected as the same value as the climb TSFC for simplicity. Lastly, the reserve loiter L/D was selected as a higher value than the converged cruise L/D because the loiter mission shoots for the best endurance and to fly at maximum L/D . The input and result values of calculating the reserve loiter mission fuel fraction are shown in Table 10.

Table 10. Reserve loiter mission fuel fraction inputs and result.

Inputs			Output
L/D	TSFC (lb/lbf/hr)	E (hr)	Reserve loiter FF
19	0.6	0.5	0.984

The fuel fractions for each of the mission profile segments as well as the total mission fuel fraction, are listed in Table 11, which also shows the starting weight at each segment (starting with the converged result of takeoff weight). In addition to the total mission fuel fraction found in Table 11, a fuel fraction of 0.005 was selected to model 0.5% of the total fuel as trapped fuel and oil (as suggested by Roskam). The two fuel fractions (the total mission fuel fraction and the fuel fraction for trapped fuel and oil) were used to calculate the total fuel weight from the takeoff weight guess. Finally, the calculated fuel weight, along with the crew weight (1,230 lb) and payload weight (57,400 lb), both specified in the requirements, the calculated empty weight is found (then converged with the allowable empty weight from the weight regression).

Table 11. Mission fuel fractions and segment start weights at the final converged MTOW.

Number	Segment	Fuel Fraction	Start Weight (lb)
1	Engine Start	0.990	322,513
2	Taxi Out	0.990	319,288
3	Takeoff	0.995	316,095
4	Climb	0.989	314,514
5	Main Cruise	0.791	311,144
6	Descent	0.990	246,115
7	Reserve Cruise	0.973	243,654
8	Reserve Loiter	0.984	237,075
9	Descent	0.990	233,282
10	Land/Taxi In	0.992	230,949
M_{ff}		0.716	

3.4 Class 1 Drag Polar Assumptions

A Class 1 drag polar is constructed to solve for the cruise L/D . The wing wetted area is calculated using an empirical relationship that takes in the takeoff weight (the guessed takeoff weight, which changes with each iteration) and regression coefficients c and d . The coefficients c and d , 0.0199 and 0.7531, respectively, are taken from Roskam's regression line coefficients for the airplane type

of transport jets.

Important assumptions made for the drag polar estimation include wing loading W_{TO}/S , C_f (equivalent skin friction coefficient), e (Oswald’s efficiency factor), and wing aspect ratio. It is important to note that wave drag is ignored as it is not relevant for the commercial airliner mission specification. The wing loading value used was 136 psf, taken from the converged value of the constraint sizing analysis presented in Section 4. Wing aspect ratio was estimated based on values from similar aircraft, where the Boeing 767-200ER and Airbus A310-300 ranged from approximately 8 to 10 for different configurations [3]. C_f was estimated as a value of 0.003 from Roskam’s plot of equivalent skin friction coefficient for jet transports. Oswald’s efficiency factor was assumed to be 0.85 as a higher-end efficiency value for commercial airliners. From these assumptions, S_{wet} , S , C_{D_0} (zero-lift drag coefficient), and K (drag polar coefficient) are solved for. These calculated values, along with the cruise velocity, cruise weight, and density, are further used to solve for the cruise lift and drag coefficients. The cruise velocity is known and the cruise weight used is taken as the average of cruise starting and ending weights (from Table 11). Lastly, the density is taken from the 1976 Standard Atmosphere for the relevant altitude for main and reserve cruise. The major assumed values for the Class 1 drag polar are shown in Table 12.

Table 12. Assumed Class 1 drag polar values.

c	d	W_{TO}/S (psf)	C_f	e	AR	Cruise ρ (slug/ft ³)	Reserve ρ (slug/ft ³)
0.0199	0.7531	136	0.003	0.85	9	0.000737	0.002048

3.5 Converged Drag Polar Results

Using the assumptions stated above, the cruise lift coefficient, drag coefficient, and L/D were calculated (and converged) and are shown for both the main cruise segment and the reserve cruise segment in Table 13.

Table 13. Main and reserve cruise lift/drag coefficients and L/D .

	C_L	C_D	L/D
Main Cruise	0.517	0.030	17.36
Reserve Cruise	0.582	0.032	18.20

Drag coefficient is calculated for a range of lift coefficients for the main cruise segment, and the resulting plot is shown in Figure 6.

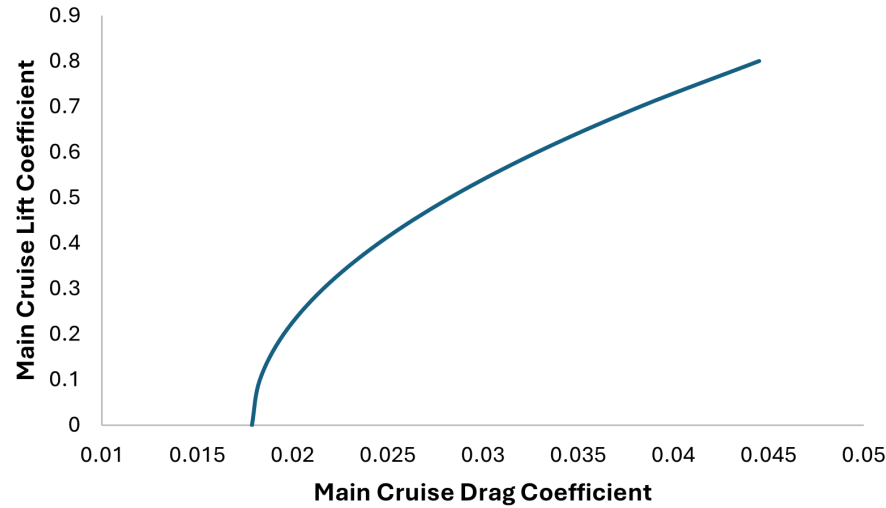


Figure 6. Main cruise segment converged drag polar.

3.6 Converged Weight Sizing Results

The converged takeoff weight result was found by iteratively solving and updating the cruise L/D value, as well as updating the takeoff weight guess value until the difference between the calculated and weight regression allowable empty weight approached zero. Table 14 shows the total fuel weight, allowable empty weight, calculated empty weight, and the converged takeoff weight.

Table 14. Converged weight sizing results at $W_{TO}/S = 136$ psf.

	Weight (lb)
Total Fuel Weight	94,812
Allowable Empty Weight	169,071
Calculated Empty Weight	169,071
Converged Takeoff Weight	322,513

It should be noted that the wing loading of 136 psf used here was not the initial Class 1 estimate, but rather the value selected from the constraint sizing analysis described in Section 4. The weight sizing was re-converged at the final wing loading so that the L/D , C_{D_0} , and resulting fuel fractions reported in this section reflect the final state of the design.

3.7 Trade Studies

Trade Study 1: Engine Technology

The cruise TSFC represents the efficiency of the propulsion system and is in an exponential term in the Breguet Range Equation. Figure 7 shows the relationship between cruise TSFC and takeoff

weight. It can be observed that a reduction in cruise TSFC results in a drastic drop in takeoff weight. As a result, investment in more efficient and modern turbofan engines is a strong method of reducing the aircraft's size.

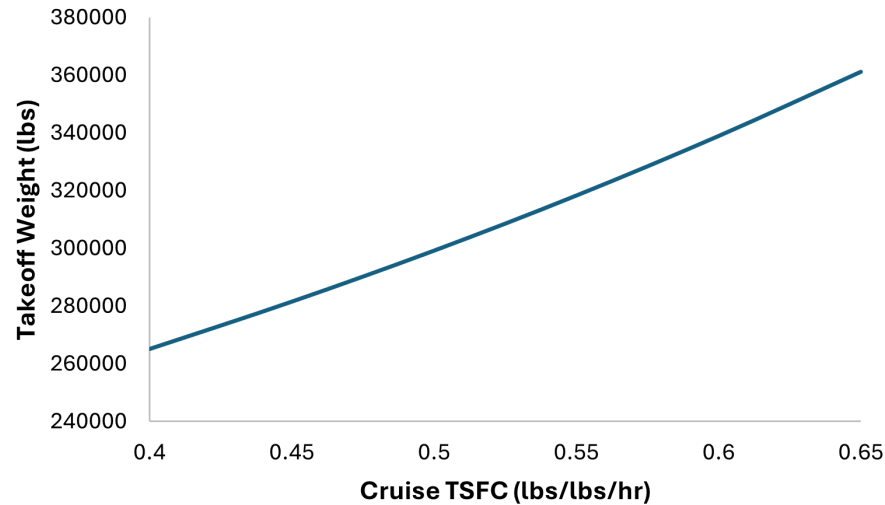


Figure 7. Takeoff weight versus cruise TSFC.

Trade Study 2: Range Mission Requirement

The mission requirement of range is the main driver of the fuel fraction. Figure 8 shows the impact of reducing and increasing cruise range on the takeoff weight. Takeoff weight increases/decreases significantly proportional to changes in the design range. The changes in range can ease the aircraft size, but also alter its marketability.

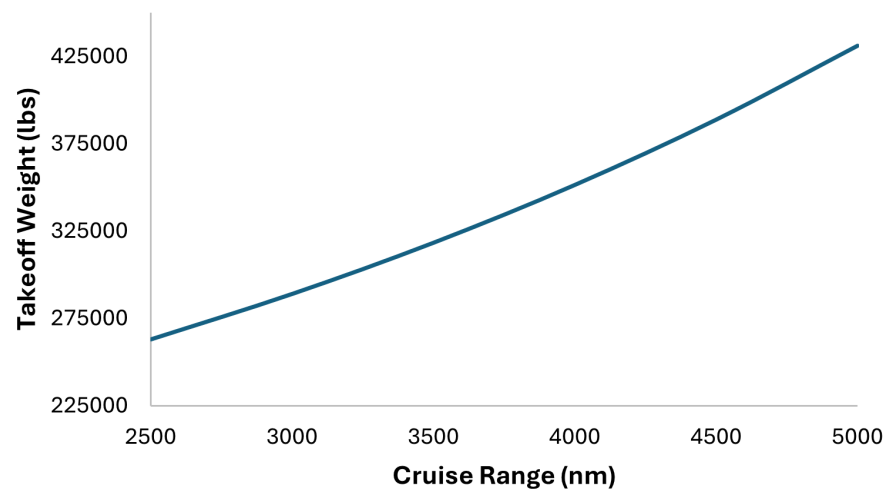


Figure 8. Takeoff weight versus cruise range.

Trade Study 3: Cruise Altitude

Varying the cruise altitude alters the air density, which is crucial in the drag polar calculation. Figure 9 shows the impact of air density and as a result the cruise altitude on the takeoff weight. It can be observed that decreases in density significantly reduce the takeoff weight, but begin to taper off around an altitude of 40,000 ft.

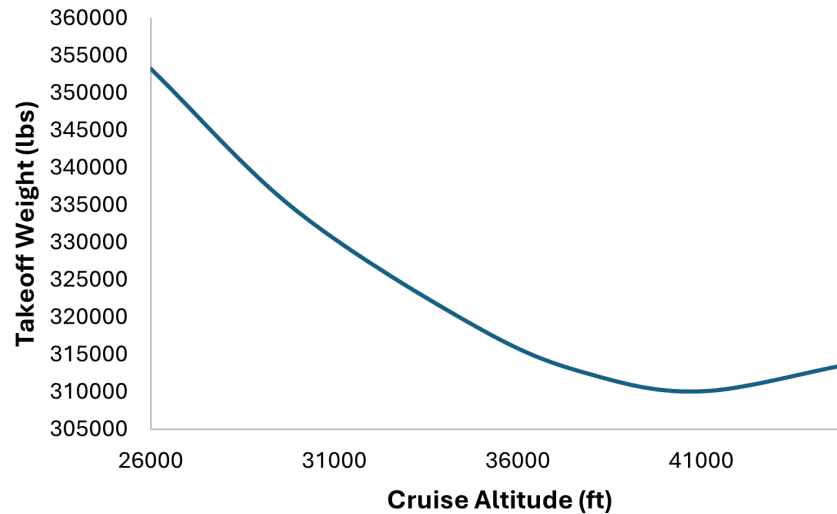


Figure 9. Takeoff weight versus cruise altitude.

3.8 Weight Sizing Conclusion

The weight sizing analysis was initially performed to produce an initial estimate of the maximum takeoff weight, and was reconverged using the wing loading of 136 psf selected from the constraint sizing analysis presented in Section 4. The final converged results are a maximum takeoff weight of 322,513 lb, an operating empty weight of 169,071 lb, and a total fuel weight of 94,812 lb, with a cruise L/D of 17.36. The trade studies demonstrated that engine efficiency, measured by cruise TSFC, has the greatest influence on takeoff weight, followed by design range and cruise altitude.

4 Constraint Sizing

4.1 Constraint Sizing Introduction

The primary goal of this section is to determine an estimate for the wing area and thrust of a middle-of-the-market commercial airliner. This is done through a constraint sizing process from an energy-based approach developed by Mattingly, Heiser, and Daley [6]. The performance constraints and requirements are set as functions of thrust loading (sea-level thrust/takeoff weight) and wing loading (takeoff weight/wing area). An energy-based constraint equation, given as Equation (1),

is derived from relations of energy balance, power, and aerodynamics. The final equation is then manipulated to solve for the constraints at specific flight conditions.

$$\frac{T_{SL}}{W_{TO}} = \frac{\beta}{\alpha} \left[\frac{q S}{\beta W_{TO}} K_1 \left(\frac{n \beta W_{TO}}{q S} \right)^2 + K_2 \left(\frac{n \beta W_{TO}}{q S} \right) + C_{D_0} + \frac{R}{q S} + \frac{1}{V} \frac{dh}{dt} + \frac{1}{g_0} \frac{dV}{dt} \right] \quad (1)$$

By solving T/W for a range of W/S values for each flight condition requirement, a constraint graph can be created, from which a final thrust-to-weight and wing loading value can be selected.

4.2 Constraints and Assumptions

The mission specification for the commercial airliner includes six flight condition requirements: max speed cruise, one engine inoperative (OEI) climb, service ceiling climb, maneuver, takeoff, and approach speed.

Some overall assumptions for the energy-based constraint equation included wing aspect ratio, zero-lift drag coefficient, Oswald's efficiency factor, and the drag coefficient of the flaps. The same assumed values from the weight sizing exercise were maintained for the wing aspect ratio and Oswald's efficiency factor of 9 and 0.85, respectively. The induced drag coefficient K is calculated from the aspect ratio and Oswald's efficiency factor as shown in Equation (2).

$$K = \frac{1}{\pi e AR} = \frac{1}{\pi \times 0.85 \times 9} = 0.0418 \quad (2)$$

The weight sizing calculated zero-lift drag of 0.0179 was used initially, and was later changed as the sizing results converged again with the final wing loading value. Finally, the flaps' drag coefficient was assumed to be 0.02 and was selected from a range of 0.015–0.02 for similar class aircraft.

Max Speed Cruise

The energy-based constraint equation can be simplified for a constant altitude/speed cruise by assuming: $dh/dt = 0$, $dV/dt = 0$, $n = 1$ (lift equals weight), and $R = 0$. The thrust lapse ratio, α , which relates the sea-level thrust to thrust at altitude, is calculated using the approximation for high-bypass-ratio turbofan engines ($M < 0.9$) given in Equation (3), from the course lecture notes.

$$\alpha = \left(0.568 + 0.25 (1.2 - M)^3 \right) \sigma^{0.6} \quad (3)$$

where $\sigma = \rho/\rho_{SL}$ is the density ratio at altitude. The Mach number used is the maximum speed requirement of 0.82, and the density is taken at the cruise altitude of 35,000 ft. The fuel/payload correction factor, β , was calculated using the starting weights of each mission segment from the weight sizing exercise. For max speed cruise, the weight before cruise was used to calculate the correction factor. Finally, velocity was calculated using the max speed requirement of Mach 0.82 at an altitude of 35,000 ft, giving 797.9 ft/s based on the 1976 Standard Atmosphere. Table 15 lists

the above-mentioned values and assumptions.

Table 15. Assumptions made for the maximum speed cruise flight condition.

α	β	Velocity (ft/s)	ρ (slug/ft ³)
0.288	0.965	797.9	0.000737

OEI Climb

The energy-based constraint equation can be simplified for a constant speed climb by assuming: $dV/dt = 0$, $n = 1$, $R = 0$. Since OEI climb occurs shortly after takeoff and is at sea level, the thrust lapse ratio is 0.5 since only one engine is active (thrust lapse ratio would equal 1 if both engines were operative). Again since OEI climb occurs shortly after takeoff, the fuel/payload correction factor was calculated using the weight at the start of climb. The velocity during OEI climb is taken as the takeoff velocity V_{TO} , calculated using Equation (4) as 1.2 times the stall speed at takeoff conditions using the maximum takeoff lift coefficient and sea-level density [6].

$$V_{TO} = k_{TO} V_{stall} = 1.2 \sqrt{\frac{2(W_{TO}/S)}{\rho_{SL} C_{L_{max,TO}}}} \quad (4)$$

Using $W_{TO}/S = 136$ psf, $\rho_{SL} = 0.002377$ slug/ft³, and $C_{L_{max,TO}} = 2.1$, this gives $V_{TO} = 280.1$ ft/s. The climb rate, dh/dt , was calculated by multiplying the velocity with the 2.4% climb gradient specified in the requirements. Additionally, a zero-lift drag coefficient increment of $\Delta C_{D_0} = 0.0025$ was added to account for the windmilling failed engine. Finally, density is assumed as the air density at sea level. Table 16 lists the above-mentioned values and assumptions.

Table 16. Assumptions made for the one-engine-inoperative climb flight condition.

α	β	Velocity (ft/s)	dh/dt (ft/s)	ΔC_{D_0}	ρ (slug/ft ³)
0.500	0.975	280.1	6.72	0.0025	0.002377

Service Ceiling Climb

The same assumptions made for the OEI climb condition can be used to simplify the energy-based constraint equation for service ceiling climb. The thrust lapse ratio was calculated using the approximation for a high-bypass turbofan engine in Equation (3) with a Mach number of 0.80 and using the air density at the service ceiling altitude of 40,000 ft. Since the service ceiling is usually approached towards the end of the cruise, the weight after the cruise segment is used to calculate the fuel/payload correction factor. The velocity is assumed from the L/D maximum point from the drag polar calculation in Section 3. The velocity was calculated by finding the lift coefficient at the L/D maximum point. Finally, the climb rate, dh/dt , was assumed to be a practical upper limit for steady flight of 100 ft/min. Table 17 shows the above-mentioned values and assumptions.

Table 17. Assumptions made for the service ceiling climb flight condition.

α	β	Velocity (ft/s)	dh/dt (ft/s)	ρ (slug/ft ³)	C_L at max L/D
0.252	0.760	773.69	1.67	0.000585	0.6694

Maneuver

The energy-based constraint equation can be simplified for a constant altitude/speed turn by assuming: $dh/dt = 0$, $dV/dt = 0$, and $R = 0$. The load factor, n , is calculated by decomposing the forces on the aircraft by the 30° bank angle requirement. The velocity is also specified by the requirements as 250 kt. The density is calculated at an altitude of 5,000 ft as specified by the requirement. The thrust lapse ratio is approximated using the high bypass turbofan engine relation with a Mach number corresponding to 250 kt at an altitude of 5,000 ft and the density of air at 5,000 ft. Finally, the fuel/payload correction factor was calculated using the weight at the end of cruise, since the aircraft typically turns after cruise to direct itself towards the arrival airport. Table 18 shows the above-mentioned values and assumptions.

Table 18. Assumptions made for the maneuver flight condition.

α	β	Velocity (ft/s)	n	ρ (slug/ft ³)
0.643	0.760	421.95	1.15	0.002048

Takeoff

The sea-level thrust to takeoff weight ratio can be approximated using the relationship for FAR 25 certified aircraft from Roskam Part I given in Equation (5), which does not assume that thrust is much greater than drag.

$$\frac{T_{SL}}{W_{TO}} = \frac{TOP_{25} (W_{TO}/S)}{C_{L_{max,TO}} \sigma S_{TO}} \quad (5)$$

where $TOP_{25} = 37.5$ is a parameter based on empirical data for FAR 25 takeoff performance, S_{TO} is the takeoff field length, and $\sigma = \rho/\rho_{SL}$ is the sea-level density ratio (equal to 1.0 at sea level). The density at sea level is used, and a takeoff field length of 9,000 ft is assumed as per the mission requirements. Finally, the maximum lift coefficient at takeoff is assumed from Roskam's typical values for maximum lift coefficients for transport jets. Table 19 shows the above-mentioned values and assumptions.

Table 19. Assumptions made for the takeoff flight condition.

TOP_{25}	Maximum C_L at takeoff	ρ (slug/ft ³)	Takeoff field length (ft)
37.5	2.1	0.002377	9,000

Approach Speed

Unlike previous flight conditions, which solve for required thrust-to-weight ratios, the approach speed requirement sets a maximum allowable wing loading. The constraint is derived from Equation (6), obtained by rearranging the stall speed equation with a FAR 25 safety factor $k_{app} = 1.3$.

$$\frac{W_{TO}}{S} = \frac{V_{app}^2 \rho C_{L_{max,L}}}{2 k_{app}^2 \beta} \quad (6)$$

The fuel/payload correction factor was calculated using the weight at the start of descent. The maximum lift coefficient at landing is assumed from Roskam's typical values for maximum lift coefficients for transport jets. The approach speed is given in the requirements of the mission as 145 kt, and finally, the density is used from sea-level altitude. Table 20 shows the above-mentioned values and assumptions.

Table 20. Assumptions made for the landing/approach flight condition.

β	Velocity (ft/s)	k_L	ρ (slug/ft ³)	Maximum C_L at landing
0.760	244.73	1.3	0.002377	2.5

4.3 Constraints Results

By calculating the sea-level thrust to takeoff weight ratio for a range of wing loading values, a constraints plot can be created, showcasing the range of allowable T/W and W/S points. Figure 10 shows the constraints plot with lines for each of the six flight conditions.

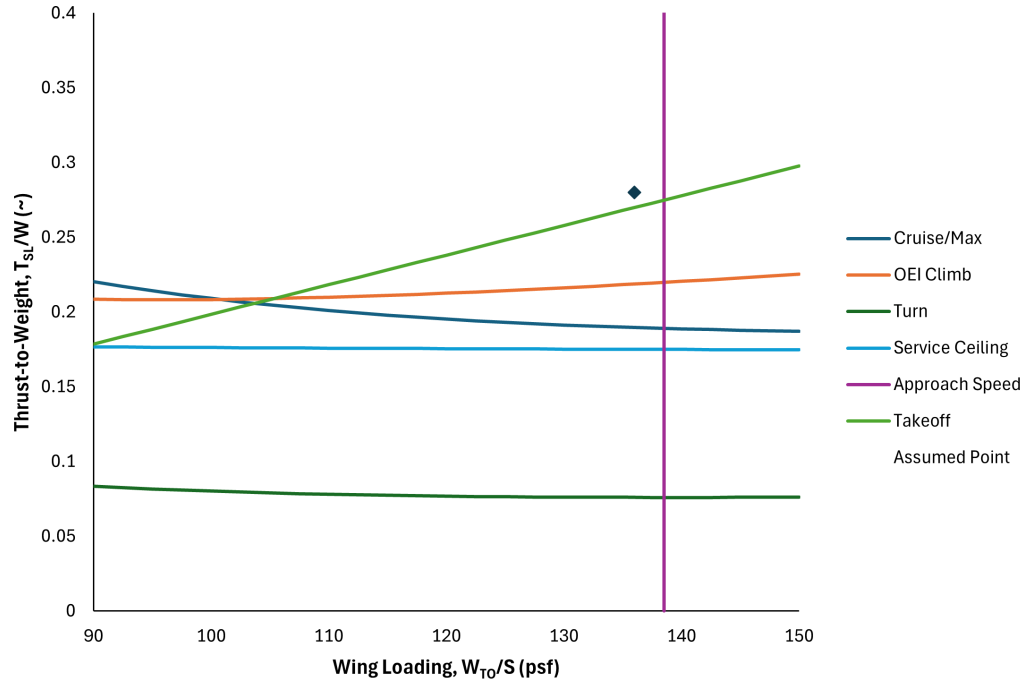


Figure 10. Sea-level thrust-to-weight versus wing loading for all six flight conditions.

The area above all the lines and to the left of the approach speed constraint in Figure 10 are valid points for T/W and W/S . This large range of acceptable values allows for the choice of optimization by either maximizing wing loading or minimizing thrust-to-weight. In this case, it was selected to maximize wing loading as it allows for a smaller wing and significantly less parasite drag. Additionally, the penalty in T/W is not as high as the penalty in W/S in the case that T/W was maximized. The assumed point in Figure 10 shows the final selected thrust-to-weight ratio and wing loading values and is shown in Table 21.

Table 21. Final selected values of T/W and W/S .

Thrust-to-Weight, T_{SL}/W_{TO}	Wing Loading, W_{TO}/S (psf)
0.28	136

The wing area and total thrust required can be calculated using the final selected ratio points from the constraints plot and the converged takeoff weight from the weight sizing presented in Section 3. These values are listed in Table 22.

Table 22. Final wing area and total thrust required.

Total Thrust (lbf)	Thrust per Engine (lbf)	Wing Area (ft ²)
90,304	45,152	2,371.4

4.4 Sensitivity Studies

Sensitivity Study 1: Takeoff Field Length

The mission requirement of 9,000 ft takeoff field length is the main constraint driver for the takeoff flight condition. Figure 11 shows the impact of reducing the takeoff field length to 8,000 ft and increasing it to 10,000 ft. Although this reduction in takeoff field length results in the need for larger and heavier engines, it could make the aircraft more accessible to smaller, more regional airports.

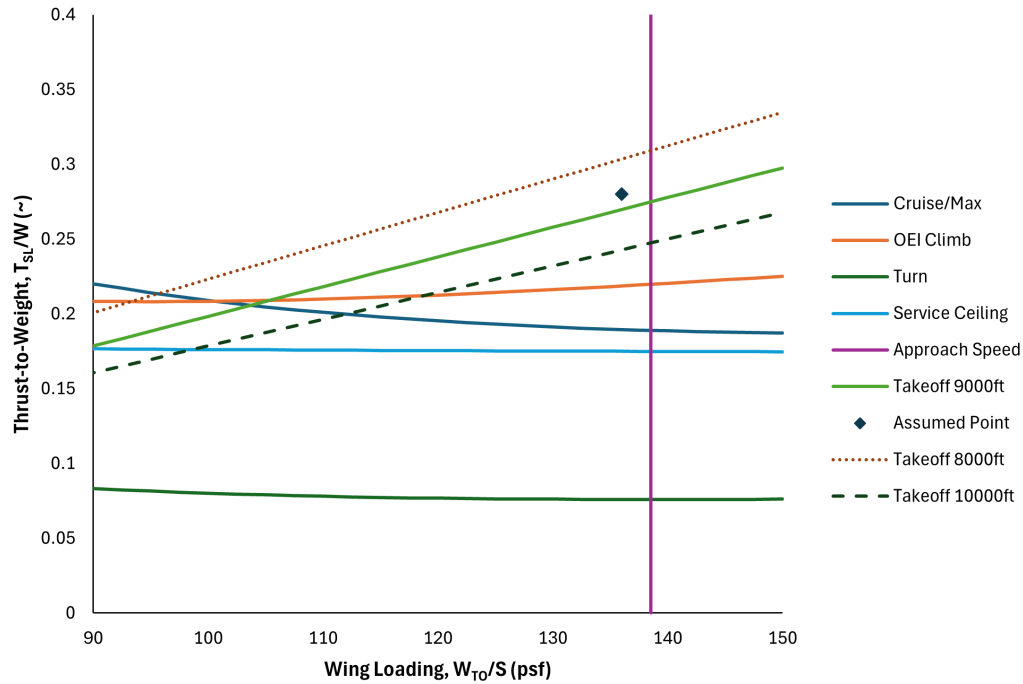


Figure 11. Constraint plot with varied takeoff field length constraints.

Sensitivity Study 2: Landing Maximum Lift Coefficient

The landing maximum lift coefficient is a main driver for the constant maximum wing loading value set by the approach speed constraint. The maximum landing lift coefficient is primarily based on the complexity of the wing flaps' aerodynamics. Figure 12 shows the effects on the constraint's plots of a higher maximum lift coefficient of 2.8 and a lower maximum lift coefficient of 2.3 (both within the recommended range by Roskam for transport jets). A higher maximum lift coefficient pushes the ceiling for wing loading, which allows for an even smaller wing.

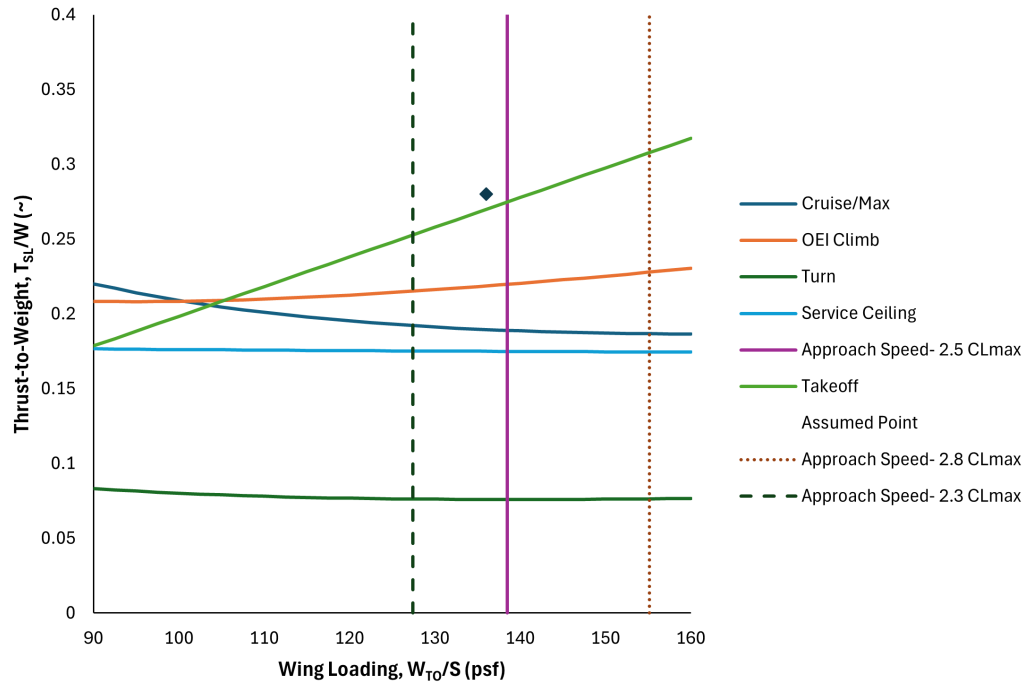


Figure 12. Constraint plot with varied maximum landing lift coefficient.

Sensitivity Study 3: Takeoff Maximum Lift Coefficient

Similar to the landing maximum lift coefficient, the takeoff maximum lift coefficient depends on the wing flaps. Heavy flaps provide significant lift but also create a large amount of drag that hurts the climb performance. If an aircraft uses fewer flaps and generates less lift, it would require a higher runway speed, meaning more powerful engines. This would cause an increase in the minimum T/W . This trend can be observed in Figure 13, which shows the impact of a lower lift coefficient of 1.8 and a higher lift coefficient of 2.2.

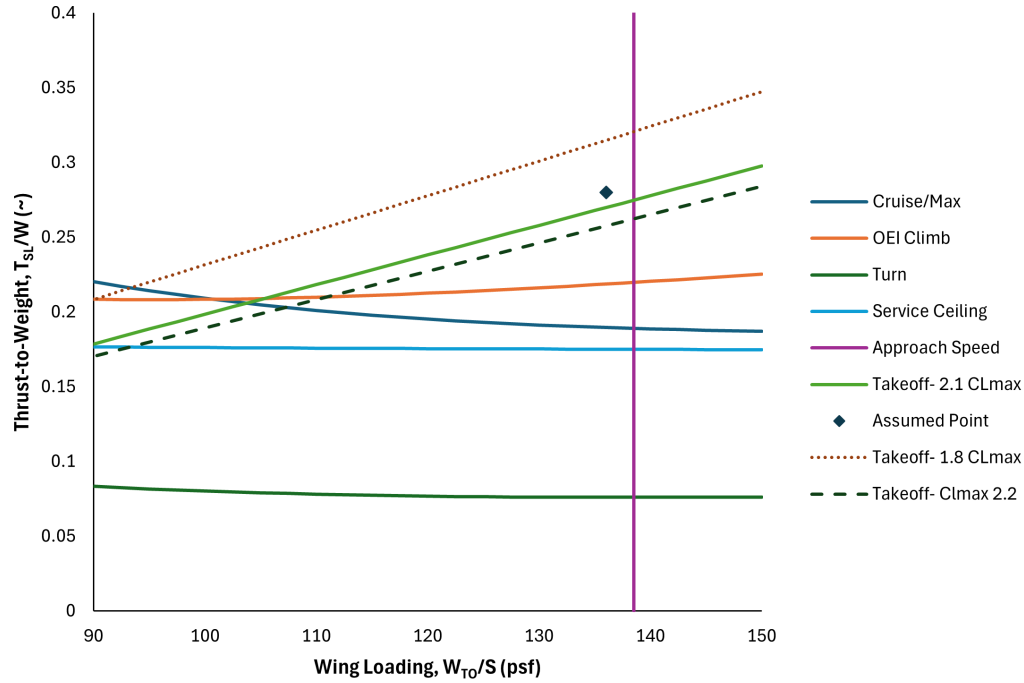


Figure 13. Constraint plot with varied maximum takeoff lift coefficient.

4.5 Weight Sizing Reconvergence

The final selected wing loading of 136 psf was used to re-converge the drag polar and weight sizing calculations from Section 3. The updated wing loading results in a new cruise L/D value from the Class 1 drag polar, which is then iterated to a new converged takeoff weight. Table 23 shows the previous (first iteration) and newly converged values for wing loading, cruise L/D , and takeoff weight.

Table 23. Previous and newly converged parameters from weight and constraint sizing.

Previous W_{TO}/S (psf)	Previous L/D	Previous W_{TO} (lb)	Previous W_E (lb)	New W_{TO}/S (psf)	New L/D	New W_{TO} (lb)	New W_E (lb)
130	17.62	318,336	167,042	136	17.36	322,513	169,071

It can be observed from Table 23 that the final takeoff weight increased after increasing the wing loading from 130 to 136 psf. This is against the expectation, as a higher wing loading should result in a smaller wing and hence a lighter aircraft. The reason for this discrepancy could arise from the calculation of the wetted wing area in the drag polar. The wing wetted area is calculated solely as a function of regression coefficients c and d and the takeoff weight, without directly accounting for the change in wing area with wing loading. As wing loading increases, the wing area decreases, which should reduce wetted area, zero-lift drag, and ultimately takeoff weight. This limitation of

the Class 1 method is noted as a source of error in the current analysis.

4.6 Constraint Sizing Conclusion

The constraint sizing analysis produced a final wing loading of 136 psf and a thrust-to-weight ratio of 0.28, selected from the design space defined by the six flight constraints. These values yield a wing area of 2,371.4 ft² and a total installed thrust of 90,304 lbf (45,152 lbf per engine) at the converged takeoff weight of 322,513 lb. The approach speed constraint was found to be the most restrictive upper bound on wing loading, while the takeoff and OEI climb drove the minimum thrust-to-weight requirement. The selected design point maximizes wing loading within the design space to minimize wing area and parasite drag. The selected wing loading was fed back into the weight sizing analysis to produce the final converged design weights reported in Section 3.

5 Cost Estimation

5.1 Cost Introduction

The primary goal of this section is to produce a cost estimate for the middle-of-the-market commercial airliner class aircraft. This is done by using cost estimating relationships (CERs) from Nicolai [2]. The development and production costs of the aircraft are considered for a range of 50–500 aircraft. Additionally, off-the-shelf items such as engine and avionics costs are considered.

5.2 Cost Drivers

The airframe acquisition cost is calculated using Nicolai’s CERs, which model the hours/cost for development and production. Both the development and production cost estimates share three main drivers: empty weight (lb), maximum speed (kt), and quantity of aircraft produced.

The empty weight was estimated in the previous weight-sizing and constraints sizing Section 3 and Section 4, and resulted in a converged allowable empty weight of 169,071 lb. The maximum speed is provided in the mission specification as Mach 0.82, and adjusted for a maximum altitude of 40,000 ft, resulting in a speed of 470 kt. The number of production aircraft was analyzed from a range of 50 to 500 aircraft. The total number of aircraft produced also includes the number of test aircraft. The number of test aircraft was selected as 4. This number is derived from FAR Part 25 certification requirements, which call for separate flight test programs for airworthiness, systems, and performance validation. Two test aircraft are used for aerodynamics and performance testing, and the remaining two are used for avionics, flight controls, and electrical systems validation. The total production quantity N for the Nicolai CERs is the sum of the production quantity and the four test aircraft.

5.3 Airframe Production Cost

The Nicolai CERs provided estimates of total hours required for the categories of: airframe engineering, tooling, manufacturing labor, and quality control. These total hour estimates are then multiplied by labor rates to provide a total cost for each task. The labor rates used in this calculation are converted to 2026 dollars and are shown in Table 24. The total cost per number of aircraft for each task is shown in 2026 dollars in Table 25. The cost conversions from 1998 dollars to 2026 dollars are made using a CPI multiplier of 1.02 from 1998–1999 and a multiplier of 1.98 [7].

Table 24. Labor rates per labor category in 2026 dollars.

Labor Category	Labor Rate (2026\$/hr)
Engineering	170.2
Tooling	174.2
Manufacturing	144.5
Quality Assurance	160.3

Table 25. Total cost per number of aircraft produced for each labor category, in 2026 dollars.

Number of A/C	Engineering (2026\$)	Tooling (2026\$)	Mfg. Labor (2026\$)	Quality Ctrl. (2026\$)
50	4,478,757,421	2,489,409,342	5,225,697,137	440,676,597
100	4,983,721,811	2,957,716,365	7,954,231,131	670,770,505
200	5,562,206,077	3,531,095,717	12,250,470,623	1,033,067,084
300	5,935,888,740	3,921,683,208	15,819,829,662	1,334,066,732
400	6,217,524,717	4,226,248,748	18,983,237,923	1,600,833,050
500	6,445,751,159	4,479,360,203	21,874,483,625	1,844,648,236

The Nicolai CERs also provided cost estimates for development support (DT&E), flight test operations, and manufacturing material and equipment. These calculations were made directly in constant and required no labor rate consideration. It is important to note that the development support and flight test operations costs do not depend on the number of aircraft produced and are therefore constant. Table 26 shows the cost estimates for development support, flight test operations, and manufacturing material and equipment.

Table 26. Total cost per aircraft for development, flight test, and manufacturing material in 2026 dollars.

Number of A/C	DT&E Support (2026\$)	Flight Test (2026\$)	Mfg. Material & Equipment (2026\$)
50	780,067,733	157,361,357	2,389,811,327
100	780,067,733	157,361,357	4,034,508,540
200	780,067,733	157,361,357	6,911,550,806
300	780,067,733	157,361,357	9,505,982,147
400	780,067,733	157,361,357	11,931,078,737
500	780,067,733	157,361,357	14,237,147,999

5.4 Other Costs

Engine and avionics costs make up other costs of the production of aircraft. The engine and avionics are assumed to be off-the-shelf items. A generic approximation for the propulsion cost can be made by the conversion of \$1,000 per lbm for turbofan engines. The weight of the GE CF6-80C2 turbofan engine is considered, as it was used for several similar-class aircraft such as the Boeing 767 and Airbus A310. The weight was found to be 9,859 lbm [8]. A generic approximation for avionics cost can be made from Raymer’s recommendation of \$4,000–\$8,000 per lbm (\$5,000/lbm was selected). The total weight of all avionics equipment was assumed to be 1,000 lb to account for all radio, management, and monitoring systems. Table 27 shows the flat cost per unit for two turbofan engines and avionics.

Table 27. Cost per unit of two turbofan engines and avionics in 2026 dollars.

Total Engine Cost (2026\$)	Avionics Cost (2026\$)
39,034,038	9,898,000

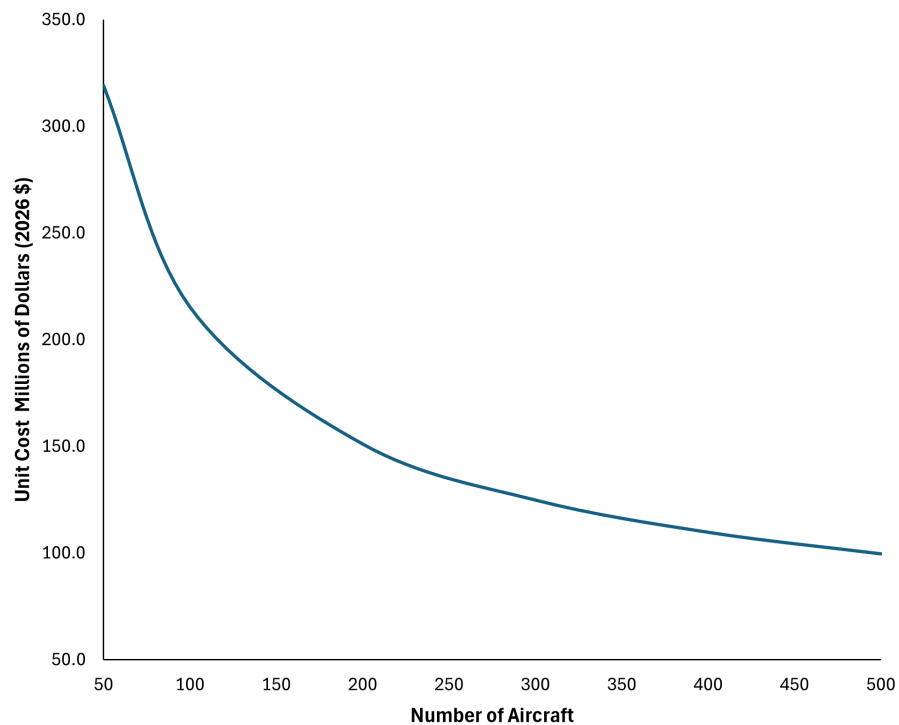
5.5 Total Cost Results

By summing up the costs from all the individual categories discussed above, one can obtain a total aircraft cost per quantity of aircraft produced. Table 28 shows the airframe total production cost, airframe cost per unit, and cost per unit of the aircraft, including the engines and avionics, per quantity of aircraft produced.

Table 28. Total and per-unit cost per quantity of aircraft produced in 2026 dollars.

Number of A/C	Airframe Total Cost (2026\$)	Airframe Unit Cost (2026\$)	Airframe + Engines + Avionics Unit Cost (2026\$)
50	15,961,780,914	319,235,618	368,167,656
100	21,538,377,443	215,383,774	264,315,812
200	30,225,819,399	151,129,097	200,061,135
300	37,454,879,579	124,849,599	173,781,636
400	43,896,352,266	109,740,881	158,672,919
500	49,818,820,313	99,637,641	148,569,678

Figure 14 shows a graph of airframe production unit cost based on Nicolai's CERs versus the number of aircraft produced. Figure 15 shows a graph of combined airframe, engine, and avionics unit cost versus the number of aircraft produced.

**Figure 14.** Airframe-only production unit cost versus number of aircraft produced.

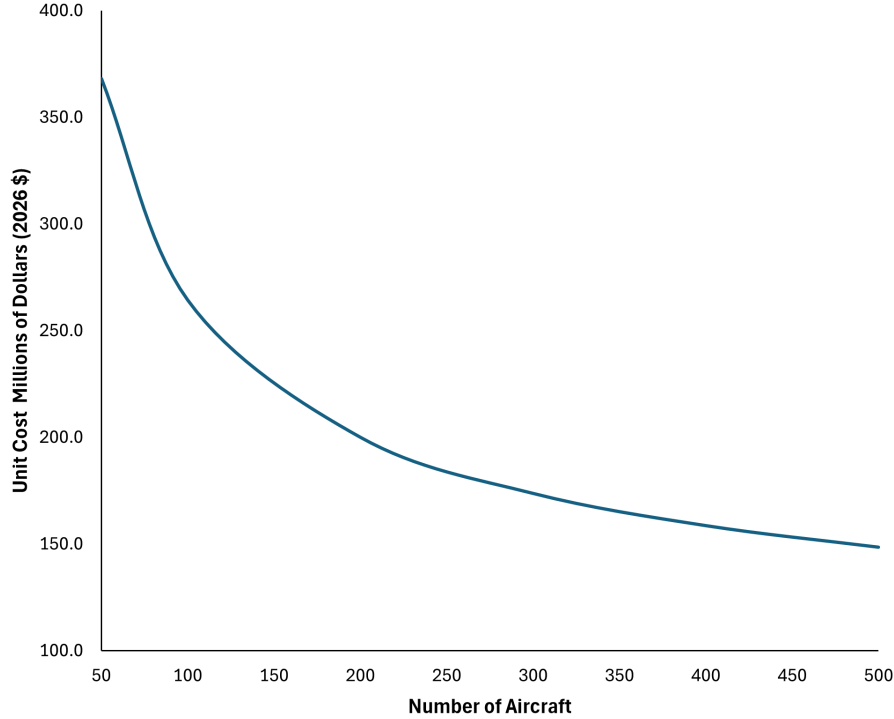


Figure 15. Combined airframe, engines, and avionics unit cost versus number of aircraft produced.

5.6 Cost Conclusion

The total unit cost of the aircraft is made up of both fixed costs and variable costs, depending on the number of aircraft produced. The Nicolai CERs model the learning curve effect, which explains the exponential nature of the plots in Figure 14 and Figure 15. It can be observed that the first 50 aircraft are much more expensive per unit than the 500th aircraft produced. The cost estimations from this model can be compared to the market price of similar aircraft; the Boeing 767-200ER has historically been listed at \$80–120 million USD. This range is lower than the cost estimation, which can be justified as the Boeing 767-200ER has been produced in much greater quantities ($\sim 1,300$), significantly reducing its unit cost through the learning curve effect.

6 Airfoil and Wing Design

6.1 Airfoil and Wing Introduction

The primary goal of this section is to select and analyze a set of airfoil candidates and determine a wing geometry for the middle-of-the-market commercial airliner. A final airfoil candidate is selected after analyzing a mix of quantitative and qualitative elements. Then, a low-speed 2D lift curve and drag polar are produced using aerodynamics analysis tools (XFOIL/XFLR5) for the final airfoil. Next, the wing geometry parameters are selected based on assumptions and similar aircraft. Finally,

3D wing analysis is performed for the selected wing/airfoil combination using the aerodynamics tool AVL, and a 3D lift curve and drag polar are generated.

6.2 Airfoil Candidate Selection

Two main factors were considered when making the selection of airfoil candidates: (1) airfoil is suitable for the transonic regime, and (2) similarity to aircraft of a similar mission specification. At the cruise speed of Mach 0.80, local airflow along the airfoil will accelerate past Mach 1.0, meaning pockets of supersonic flow. To mitigate the shock effect that may occur, airfoils of supercritical shape are used. Supercritical airfoils typically have a flattened upper surface to prevent the flow from accelerating violently, and a highly cambered trailing edge to generate extra lift sacrificed from a flatter top surface. These factors were considered, and the following airfoils were selected:

- B737a – Boeing 737 mid-span
- BACXXX – Boeing Energy Efficient Transport Program airfoil
- NASA SC(2)-0714
- NASA SC(2)-1010

The B737a airfoil was selected as it is one of the airfoil shapes used in the Boeing 737 aircraft. The Boeing 737 is one of the most prevalent commercial transport jet aircraft which aligns very closely with the mission specifications. This provides a strong baseline candidate that has proven successful for similar missions. Similarly, the BACXXX airfoil was selected as it was designed for similar transport missions. Additionally, similar airfoils to the BACXXX were demonstrated with Boeing 757/767 aircraft with the Boeing Energy Efficient Transport Program.

The final two airfoil candidates are both NASA supercritical airfoils of varying design lift coefficient and thickness-to-chord ratio as a percentage. The NASA SC(2)-0714 airfoil has a design lift coefficient of 0.7 and a maximum thickness of 14%. The high thickness allows for a greater fuel volume while maintaining an adequate designed lift coefficient. The NASA SC(2)-1010 has a design lift coefficient of 1.0 and a maximum thickness of 10%. This candidate provides a test if aggressively maximizing lift is worth it when comparing all the airfoils in the down-selection process.

6.3 Airfoil Down-Selection

The airfoil down-selection is performed using a decision matrix with a mix of quantitative and qualitative metrics.

The main quantitative metrics include maximum lift coefficient, maximum lift-to-drag ratio (L/D), pitching moment coefficient (at cruise, around lift coefficient of 0.6), and thickness-to-chord ratio (t/c). The highest maximum lift coefficient and the highest maximum L/D are desirable as they provide the greatest aerodynamic performance. A pitching moment coefficient closer to zero (less negative) is more desirable as it reduces the nose-down twisting force felt by the aircraft

and the complexity required of the tail wings. Lastly, a greater t/c value is more desirable as it maximizes the internal volume available for fuel, despite potential increases in profile drag.

The qualitative metrics include stall behavior, transonic suitability, and manufacturability. A smoother round-off stall behavior is more desirable than a case with lift coefficient dropping abruptly at stall. Transonic suitability measures the airfoil shape's ability to delay wave drag. Finally, manufacturability considers the thickness and sharp edges of the airfoil.

To assist the down-selection process, section lift coefficient, section drag coefficient, and section moment coefficient data for the four airfoil candidates are generated using XFLR5 at conditions of 5×10^6 Reynolds number and 0.0 Mach number. Figure 16 shows the 2D lift curves for the four airfoil candidates, and Figure 17 shows the 2D drag polars for the four airfoil candidates.

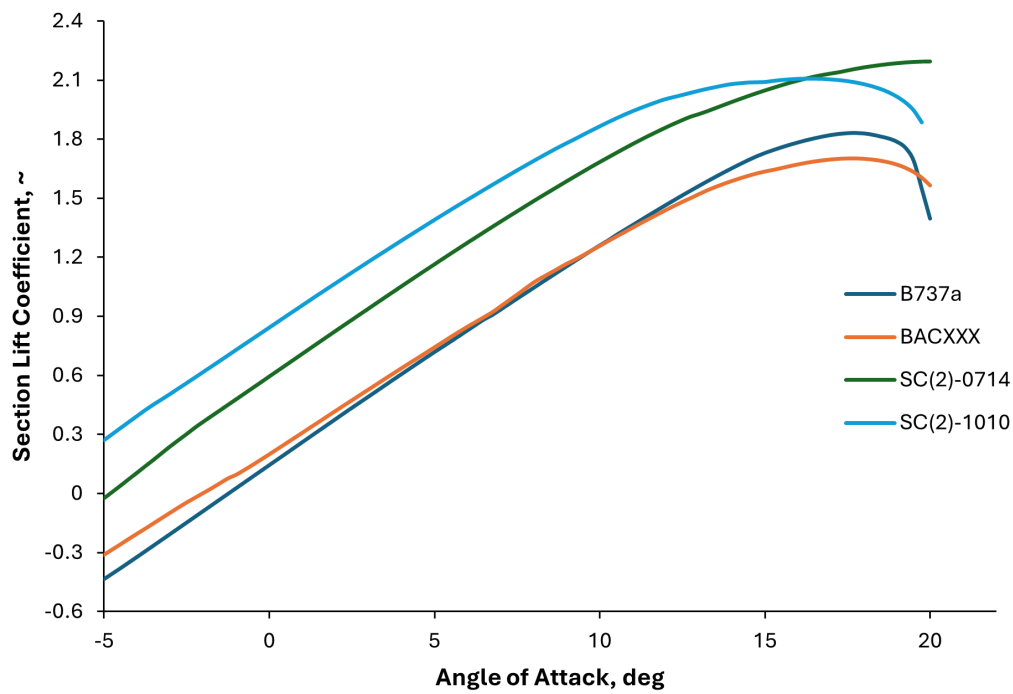


Figure 16. Section lift coefficient versus angle of attack for the four airfoil candidates.

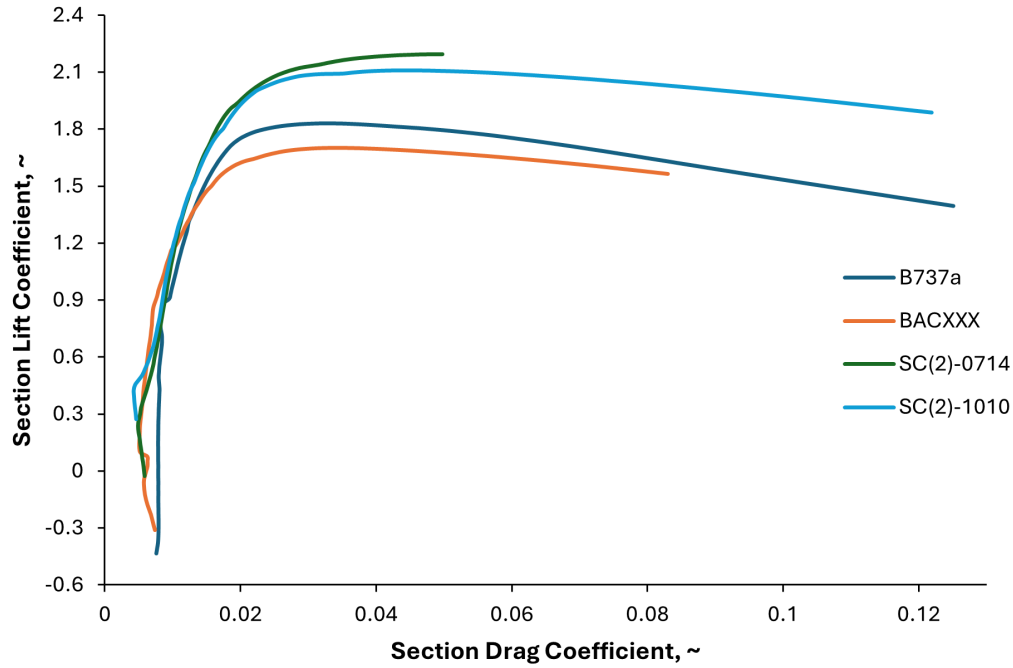


Figure 17. 2D drag polar plot for the four airfoil candidates.

Using the above-mentioned criteria, the candidate airfoils were ranked for each parameter. Table 29 shows the decision matrix and the score for each category, as well as the final selected airfoil.

Table 29. Decision matrix for the four candidate airfoils.

Parameter	Airfoil Value				Score			
	B737a	BACXXX	SC(2)-0714	SC(2)-1010	B737a	BACXXX	SC(2)-0714	SC(2)-1010
Maximum C_L	1.83	1.70	2.19	2.11	2	1	4	3
Maximum L/D	105.78	120.61	117.01	118.66	1	4	2	3
Cruise C_m	-0.002	-0.039	-0.144	-0.152	4	3	2	1
t/c	15.4%	11.3%	14.0%	10.0%	4	2	3	1
C_L at $\alpha = 0^\circ$	0.14	0.20	0.59	0.84	1	2	3	4
C_L at min C_D	0.14	0.15	0.24	0.42	1	2	3	4
Stall behavior	—	—	—	—	1	3	4	2
Transonic suitability	—	—	—	—	1	2	4	3
Manufacturability	—	—	—	—	4	2	3	1
Sum					19	21	28	22

The final selected airfoil is the NASA SC(2)-0714. The shape of the selected airfoil is shown in Figure 18, and the characteristic flattened upper surface and aft-cambered trailing edge of supercritical airfoils can be observed. The 2D lift curve for the NASA SC(2)-0714 airfoil is shown in Figure 19, and the 2D drag polar for the NASA SC(2)-0714 airfoil is shown in Figure 20 (with data

from XFLR5 at Mach 0.0 and 5×10^6 Reynolds number, using coordinates from the UIUC airfoil database [9]).

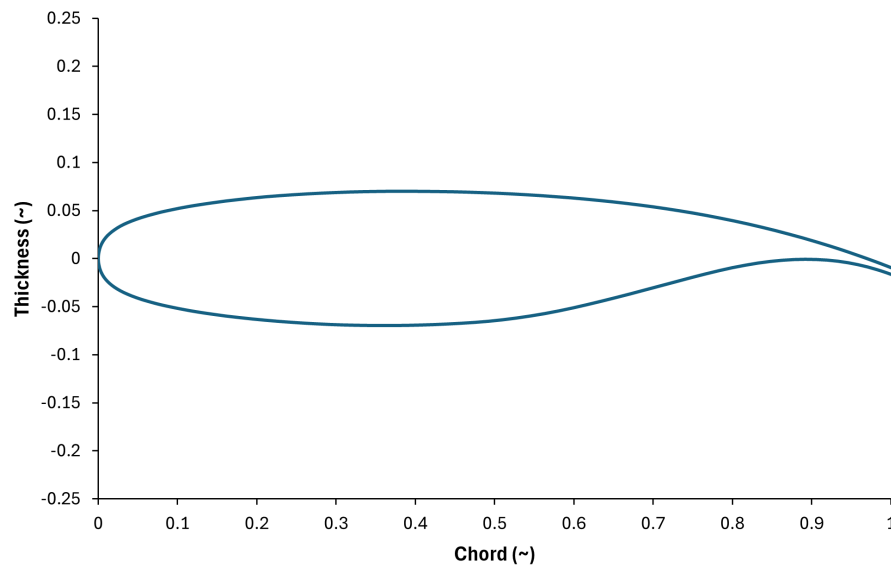


Figure 18. NASA SC(2)-0714 airfoil shape.

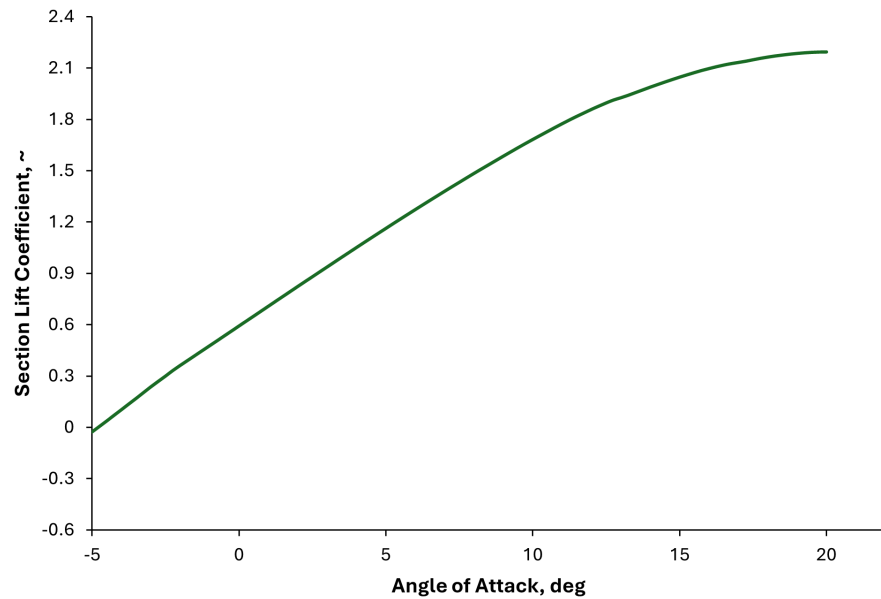


Figure 19. Section lift coefficient versus angle of attack for the NASA SC(2)-0714 airfoil.

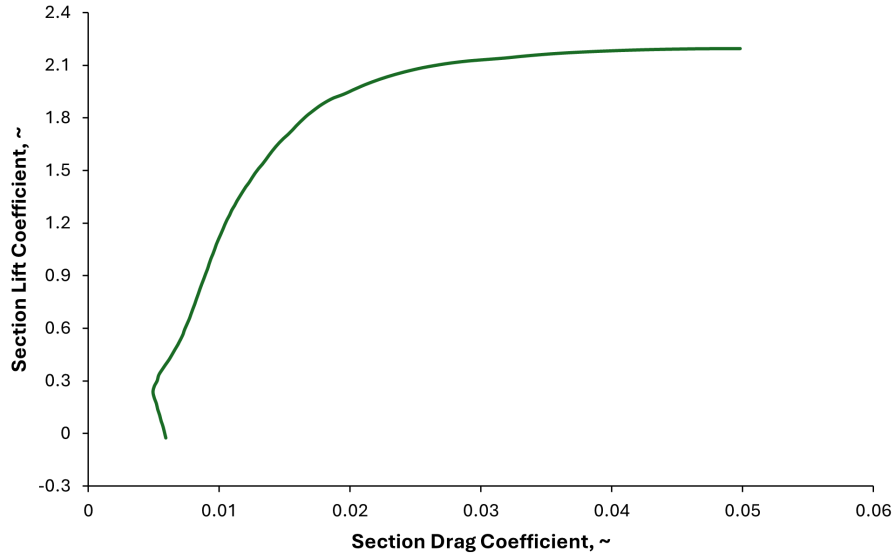


Figure 20. 2D drag polar plot for the NASA SC(2)-0714 airfoil.

6.4 3D Wing Geometry Selection

The wing geometry parameters were selected to model a simple tapered wing as it provides a successful design for commercial airliners that balances performance and manufacturing ease. An aspect ratio of 9 was assumed for previous exercises based on similar aircraft such as the Boeing 767-200 and the Airbus A310 [3], so this assumption was kept. The previously performed constraint sizing yielded a wing loading of 136 psf and a wing area of 2,371.4 ft². Based on similar aircraft to the mission specification, the following assumptions were made: sweep angle (at the quarter chord) of 30°, taper ratio of 0.3, dihedral of 4°, and a twist of −2°. Using these values along with the converged wing area, the wingspan and root/tip chord lengths were calculated. These values and assumptions are listed in Table 30.

Table 30. Assumed and previously converged parameters for the 3D wing design.

AR	Sweep at c/4 (deg)	Taper ratio	Dihedral (deg)	Twist (deg)	Wing area (ft ²)	Wingspan (ft)	Root chord (ft)	Tip chord (ft)
9	30	0.3	4	−2	2,371.4	146.09	24.97	7.49

A view of the resulting wing geometry, generated in AVL, is shown in Figure 21.

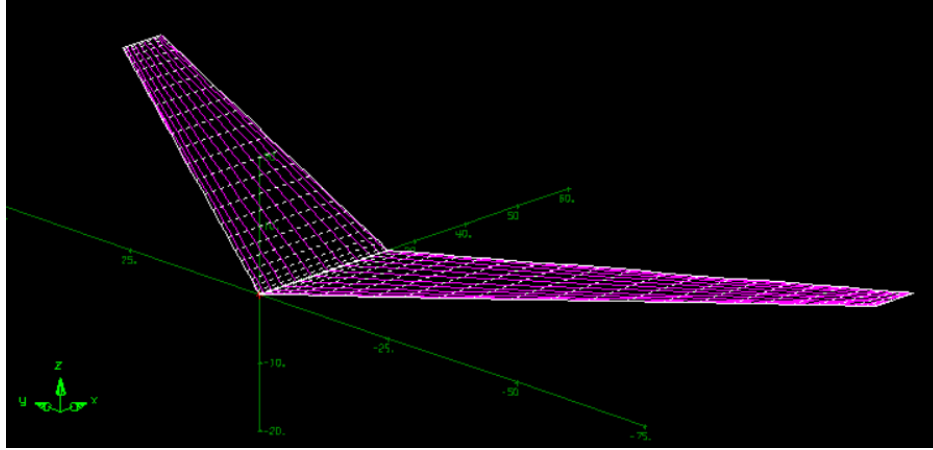


Figure 21. View of the 3D wing geometry generated in AVL.

The selection of a 30° sweep angle was further verified through a drag divergence Mach number trade study using the Korn equation [2] given in Equation (7).

$$M_{dd} = \frac{\kappa_A}{\cos \Lambda} - \frac{t/c}{\cos^2 \Lambda} - \frac{C_L}{10 \cos^3 \Lambda} \quad (7)$$

The airfoil technology factor κ_A is taken as 0.95 for supercritical airfoils, with $t/c = 0.14$ and a cruise lift coefficient of 0.517. Table 31 shows the resulting drag divergence Mach number for sweep angles between 0° and 45° .

Table 31. Drag divergence Mach number versus quarter-chord sweep angle.

Sweep angle (deg)	M_{dd}
0	0.758
10	0.766
15	0.776
20	0.790
25	0.808
30	0.831
35	0.857
40	0.887
45	0.917

The cruise Mach number requirement of 0.80 must be below the drag divergence Mach number to avoid a sharp rise in compressibility drag. From Table 31, sweep angles below approximately 25° result in M_{dd} below the cruise Mach. The selected sweep of 30° yields $M_{dd} = 0.831$, providing

margin above the cruise Mach. Higher sweep angles would provide additional margin but may come with structural weight penalties and reduced low-speed lift, making 30° a reasonable balance for this design.

Using these parameters, a 3D wing analysis was able to be performed for the wing and airfoil combination. Figure 22 shows the lift curve, and Figure 23 shows the drag polar from the 3D wing analysis done on AVL. As AVL only estimates induced drag, the 3D drag polar is labeled as such.

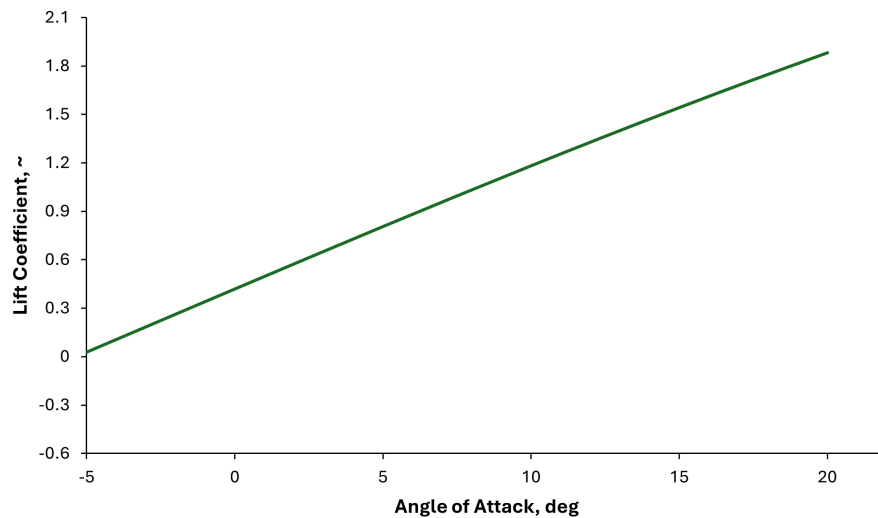


Figure 22. Lift curve for the 3D wing and airfoil design from AVL.

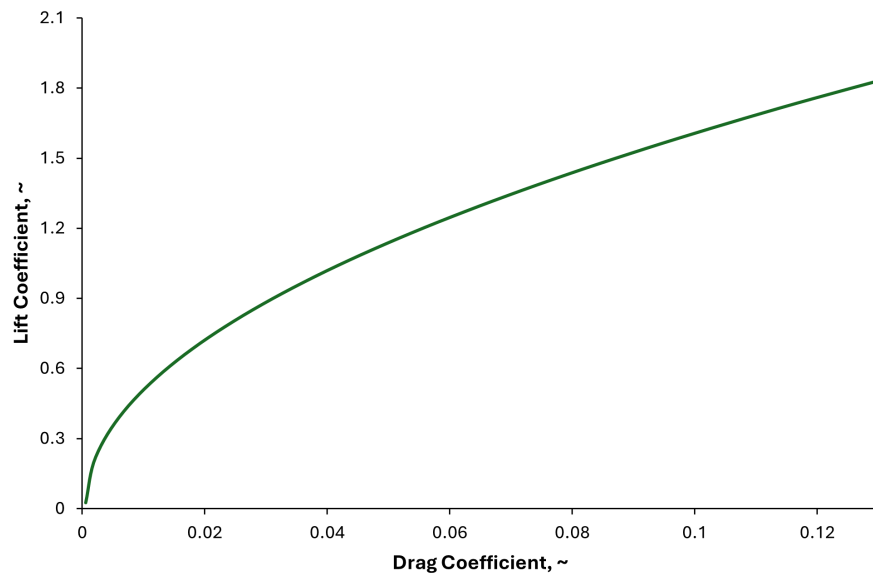


Figure 23. Induced-drag polar for the 3D wing and airfoil design from AVL.

6.5 Airfoil and Wing Conclusion

The airfoil and wing analysis selected the NASA SC(2)-0714 supercritical airfoil based on a weighted decision matrix scoring various quantitative and qualitative factors. The 3D wing was sized using the converged wing area of $2,371.4 \text{ ft}^2$ and wing loading of 136 psf from the constraint sizing analysis, with an aspect ratio of 9, a quarter-chord sweep of 30° , taper ratio of 0.3, dihedral of 4° , and a twist of -2° . The resulting wingspan is 146.09 ft, with a root chord of 24.97 ft and a tip chord of 7.49 ft. The drag divergence Mach number trade study using the Korn equation confirmed that the selected sweep provides suitable margin above the cruise Mach of 0.80. 3D AVL analysis produced a lift curve and induced drag polar for the wing design. The parameters established in this section feed into the subsequent fuselage and tail sizing analysis presented in Section 7.

7 Fuselage and Tail Sizing

7.1 Fuselage and Tail Sizing Introduction

The primary goal of this section is to determine the sizing of the fuselage and tail surfaces for a middle-of-the-market commercial airliner. The first part includes a basic fuselage sizing. This is done by defining all fuselage requirements and defining the maximum cross-sectional size and other key geometrical parameters. Next, the tail sizing is performed based on the volume coefficient method. The center of gravity location is estimated and allows for the calculation of the tail area. Finally, the lifting surface parameters are all defined similarly to the wing geometry. The total configuration of the wing, fuselage, and tail is visualized using OpenVSP.

7.2 Fuselage Sizing

In determining the fuselage sizing, it is important to start with the requirements specified in the mission specifications. The requirements state there are two pilots, four flight attendants, and a maximum of 280 passengers, each at 175 lb and having 30 lb of baggage. The cabin layout will consist of a single class arrangement with galleys, lavatories, and wardrobes. To meet the services of 280 passengers during a 3,500 nm mission, the fuselage will accommodate two galleys. The total number of lavatories for the fuselage was selected as six, which follows an approximation of one lavatory to every 50 passengers. The fuselage shall contain two wardrobes to contain crew equipment and other passenger storage.

The seating arrangement is modeled with the goal of minimizing the number of seat rows to accommodate the lavatory/galley sections without an excessively narrow fuselage. As a result, the number of seats abreast was selected as eight, with two aisles meeting the FAR 25 size requirement with a width of 19 in. The seat pitch is selected for a high-density arrangement, as all the seating will be economy class, and was selected as 31 in. Other assumptions made contributing to the maximum cross-sectional size include a seat width of 18 in, an armrest width of 2 in, and a wall thickness of 6 in, which adhere to typical commercial practices. Summing up all the widths results

in a diameter of 216 in: 18×8 (seat width) + 2×11 (armrest width) + 19×2 (aisle width) + 6×2 (wall thickness) = 216 in or 18 ft.

With the fuselage diameter (d_f) defined at 18 ft, the remaining key geometric parameters can be assumed. The fineness ratio (fuselage length/fuselage diameter) was selected as 10.5 out of a range of currently used geometric fuselage parameters of 6.8–11.5 for transport jets provided by Roskam. A value on the higher end of the range was selected as a longer, skinnier shape minimizes pressure drag, which is important at a cruise speed of Mach 0.8, despite an increase in skin friction drag. The higher fineness ratio also allows for sufficient nose and tail cone lengths. A tailcone ratio (fuselage tail length/fuselage diameter) of 3 was selected from a range of currently used geometric fuselage parameters of 2.6–4 for transport jets provided by Roskam. This value was selected as it provides sufficient tail section length to prevent aft flow separation, but also to allow for space for the nose cone (further explained below).

The two selected ratio values and the determined fuselage diameter allow for the calculation of the total fuselage length and the tailcone length. This results in a fuselage length (l_f) of 189 ft and a tailcone length (l_{fc}) of 54 ft. The nose cone length can be determined by subtracting the tailcone length and the cabin length from the total fuselage length. Cabin length can be estimated using the selected seat pitch and approximations for the length of the forward, middle, and aft lavatory/galley/wardrobe sections. For the arrangement of 280 passengers, eight seats abreast, and 31 in seat pitch, there will be 35 rows, resulting in a total of 90.4 ft for the seating sections. The forward and aft cabin provisions will include two lavatories, one galley, and one wardrobe, and are estimated to cover a length of 10 ft each. The middle cabin provisions will include two lavatories and are estimated to cover a length of 4 ft. This results in a total cabin length of 114.4 ft. The nose cone length can then be calculated and is determined as 20.6 ft.

The last key external fuselage geometric parameter is the aft upsweep angle. Roskam provided a range of currently used geometric fuselage parameters of 11 – 16° for transport jets. Selecting the upsweep angle is a balance of performance and safety. A smaller angle is more ideal as it minimizes pressure drag, but it restricts the maximum rotation angle during takeoff. An upsweep angle of 12° is selected to maintain aerodynamic performance, and since the selected V-tail configuration should not be at risk of contact with the ground. The key fuselage sizing parameters discussed above are defined in Table 32.

Table 32. Key fuselage shape sizing parameters.

Fuselage diameter (ft)	Fineness ratio	Tailcone ratio	Aft upsweep angle (deg)	Cabin length (ft)	Tailcone length (ft)	Nose cone length (ft)	Total fuselage length (ft)
18	10.5	3	12	90.4	54	20.6	189

The parameters specified in Table 32 can be used to draw the full fuselage shape. Figure 24 shows the fuselage drawn using OpenVSP.

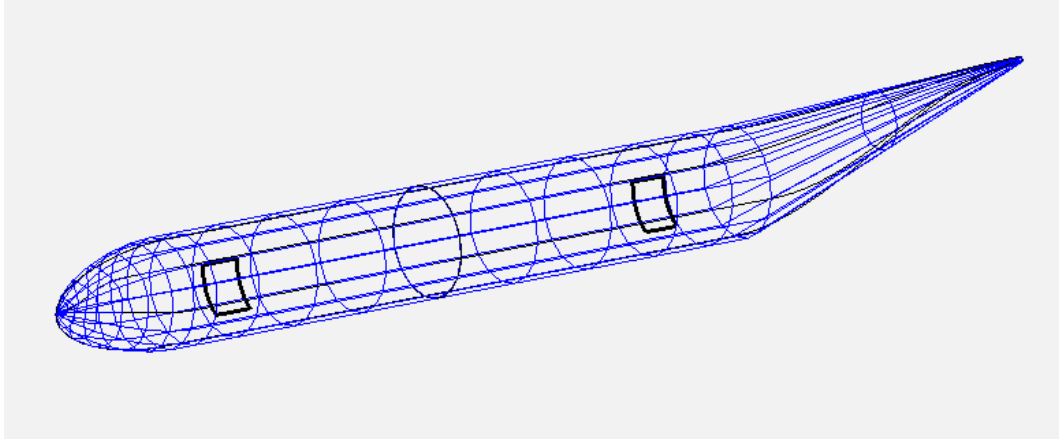


Figure 24. Fuselage with parameters from Table 32 visualized using OpenVSP.

In order to determine the floor height of the fuselage cabin cross-section, the type of cargo compartment must be decided. The wide 18 ft diameter of the fuselage allows for the fit of two LD3 cargo pallets side by side. LD3 is a standardized cargo pallet for commercial transports and provides an optimized fit storage compartment for the bottom of the fuselage. The compartment's height and base width, along with a structural depth of 6 in (rounded up from calculating $0.02d_f + 1$ in for large transports) determined a floor height of 99 in or 8.25 ft. This provides a height of 9.75 ft for the passenger level of the cabin, which is sufficient for walking and overhead cabin storage spaces. Figure 25 shows a sketch of the fuselage cross-section drawn to scale with two LD3 cargo compartments. Figure 26 shows a sketch of the fuselage cross-section drawn to scale with seats, aisles, and overhead bin space according to the width assumptions mentioned above (seat width of 18 in, armrest width of 2 in, aisle width of 19 in). The height from the floor to the overhead storage structure is 82.5 inches.

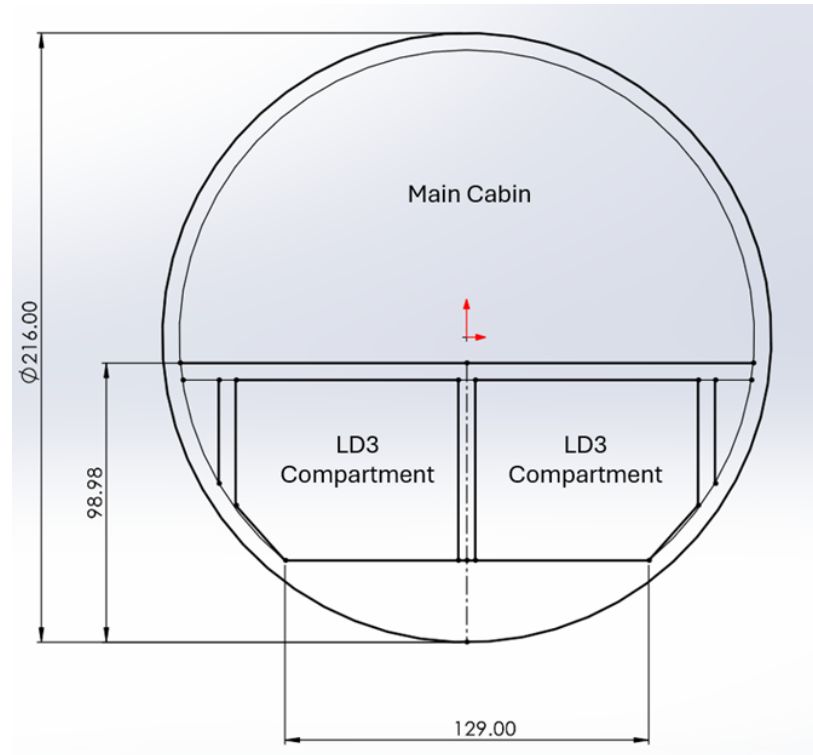


Figure 25. Cabin and cargo cross-section including two LD3 storage compartments.

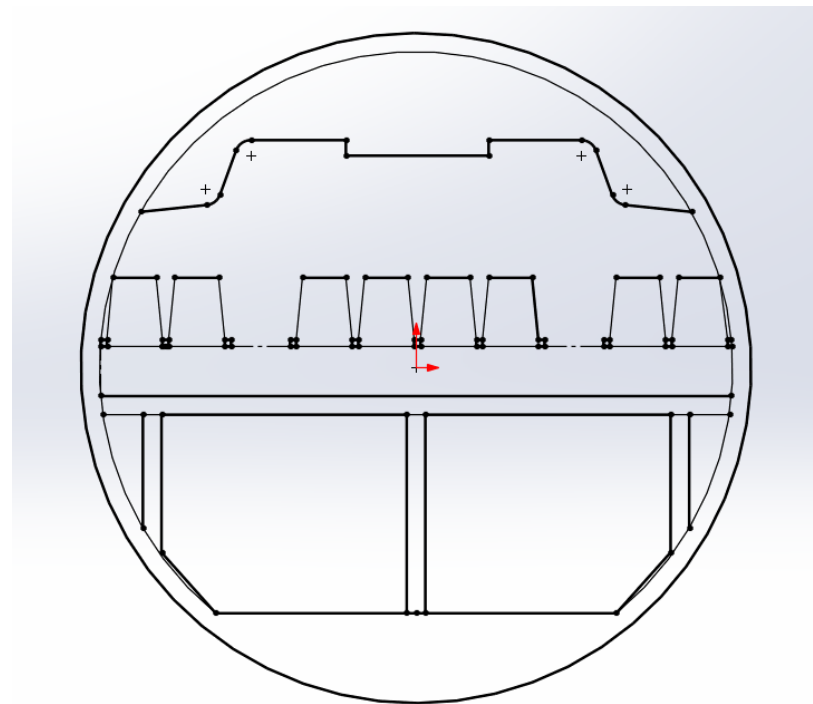


Figure 26. Cabin and cargo cross-section including seats, aisles, and overhead bin space.

7.3 Tail Sizing

The tail configuration for this aircraft was previously selected as V-tail, and the sizing is determined based on the volume coefficient method. The horizontal and vertical tail coefficients are assumed based on a similar aircraft of the Boeing 767-200, which meets similar requirements and sizing to the middle-of-the-market commercial airliner being designed. The vertical tail coefficient used is 0.067, and the horizontal tail coefficient is assumed to be 1. The vertical and horizontal tail area is calculated using the volume coefficients, wing area, mean aerodynamic chord, and wingspan. The wing area and wingspan determined in previous exercises are 2,371.42 ft² and 146.09 ft, respectively. The mean aerodynamic chord is calculated using Equation (8), which uses the taper ratio of 0.3 and the root chord of 24.97 ft from previous exercises.

$$\bar{c} = \frac{2c_{\text{root}}}{3} \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right) \quad (8)$$

This resulted in a mean aerodynamic chord of 17.8 ft. The vertical and horizontal tail areas were calculated using an estimated moment arm of 85 ft from the center of gravity for both the vertical and horizontal tails in Equations (9) and (10).

$$S_v = \frac{\bar{V}_v S b}{x_v} \quad (9)$$

$$S_h = \frac{\bar{V}_h S \bar{c}}{x_h} \quad (10)$$

The dihedral angle for the V-tail can then be calculated using the vertical and horizontal tail areas in Equation (11). The total V-tail area can be calculated using the horizontal tail area (as it is more dominant) and the dihedral angle using Equation (12).

$$\Gamma_h = \arctan\left(\frac{S_v}{S_h}\right) \quad (11)$$

$$S_{v\text{-tail}} = \frac{S_h}{\cos^2(\Gamma)} \quad (12)$$

The resulting values are defined in Table 33.

Table 33. V-tail sizing parameters determined from the volume coefficient method.

Vertical tail volume coefficient	Horizontal tail volume coefficient	Vertical tail area (ft ²)	Horizontal tail area (ft ²)	Dihedral (deg)	V-tail total area (ft ²)
0.067	1	496.6	273.1	36.5	768.5

With the V-tail area determined, the lifting surface parameters can be defined. Despite the V-tail having both a horizontal and vertical component, the ranges provided by Roskam for horizontal tail design parameters are used since the horizontal component is more dominant. The aspect ratio is

assumed to be 4.5, which is half of the wing aspect ratio. The sweep angle (at quarter chord) is assumed to be 35° , which is 5° greater than the wing sweep angle. This accounts for the tail's higher critical Mach number. The taper ratio is assumed to be 0.4 to balance the structural weight and manufacturing simplicity. The NACA 0012 airfoil is selected for the V-tail as it is a symmetrical airfoil. Airfoil symmetry in the tail wings allows for a simple control force in both the positive and negative directions. Table 34 shows all the assumed values for the V-tail.

Table 34. V-tail lifting surface parameter assumptions.

Total tail area (ft ²)	Aspect ratio	Sweep angle at quarter chord (deg)	Taper ratio	Dihedral (deg)	Twist (deg)
768.5	4.5	35	0.4	36.5	0

With the fuselage sizing parameters, tail sizing parameters, and wing sizing parameters from the previous exercise, a total configuration is drawn in OpenVSP and is shown in Figure 27.

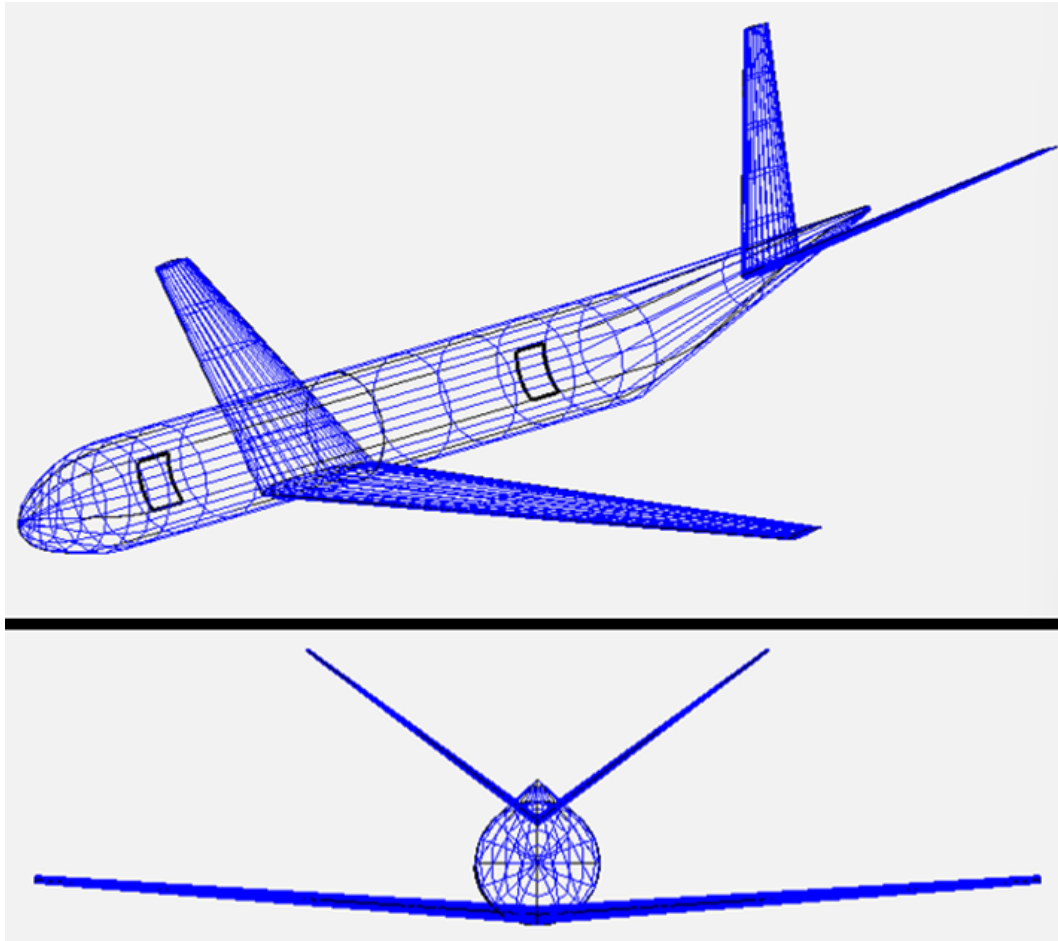


Figure 27. Total configuration of wing, fuselage, and tail drawn in OpenVSP.

7.4 Fuselage and Tail Sizing Conclusion

The fuselage was sized to accommodate 280 passengers in an eight-abreast, single-class arrangement with two aisles. This led to a fuselage diameter of 18 ft and a total length of 189 ft. The fineness ratio of 10.5 and tailcone ratio of 3 were selected to minimize pressure drag while providing adequate nose and tail cone lengths. Two LD3 cargo pallets fit side by side in the lower fuselage, resulting in a cabin floor height of 8.25 ft and a passenger cabin height of 9.75 ft. The V-tail was sized using the volume coefficient method, with a vertical tail volume coefficient of 0.067 and a horizontal tail volume coefficient of 1.0, producing a total V-tail area of 768.5 ft² at a dihedral angle of 36.5°. The total configuration of the fuselage, wing, and V-tail is shown in Figure 27 and serves as the geometric basis for the weight and balance analysis presented in Section 8.

8 Weight and Balance

8.1 Weight and Balance Introduction

The primary goal of this section is to identify the weight of the high-level components of a middle-of-the-market commercial airliner and determine the longitudinal center of gravity (CG). This is performed using the empty weight calculation from the previous sizing sections, as well as historical weight ratios for similar aircraft from Roskam Part V [10]. From there, an approximate CG location is defined for each of the components. The component CG locations allowed for a complete weight and balance table for the longitudinal axis, and the calculation of CG location variation for different fuel/payload layouts. Lastly, a CG excursion diagram is created, and the landing gear load distribution is verified to meet requirements.

8.2 Component Weight and Center of Gravity Location Estimation

To find the weights of the high-level components in the aircraft, historical gross weight ratios for four similar aircraft (Boeing 737-200, Boeing 727-100, Boeing 747-100, and Airbus A-300 B2), taken from Roskam Part V, are averaged. Table 35 shows the gross weight (GW) ratios for each of the similar aircraft, and their average.

Table 35. Historical gross weight ratios for transport jet aircraft.

Aircraft	Boeing 737-200	Boeing 727-100	Boeing 747-100	Airbus A300-B2	Average
Gross Weight (lb)	115,500	160,000	710,000	302,000	321,875
Wing/GW	0.092	0.111	0.122	0.146	0.118
Empennage/GW	0.024	0.026	0.017	0.020	0.022
Fuselage/GW	0.105	0.111	0.101	0.119	0.109
Nacelle/GW	0.012	0.024	0.014	0.023	0.018
Landing Gear/GW	0.038	0.045	0.044	0.045	0.043
Propulsion/GW	0.071	0.078	0.062	0.076	0.072
Fixed Equip/GW	0.129	0.133	0.089	0.116	0.117

The average gross weight ratios were used along with the previously determined empty weight of 169,071 lb. The converged weights from the weight and constraint sizing sections, which serve as the basis for the component weight estimation, are summarized in Table 36.

Table 36. Top-level weight breakdown from weight sizing.

Weight Item	Weight (lb)
Maximum Takeoff Weight (W_{TO})	322,513
Operating Empty Weight (W_{OE})	169,071
Total Fuel Weight (W_f)	94,812
Payload Weight (W_{PL})	57,400
Crew Weight (W_{crew})	1,230

The weight of each component was proportionally adjusted to ensure that the estimated empty weight equals that of the previously determined value. Table 37 shows the component weight estimates, their adjustments, and the final weight of each component. Table 37 also shows the required crew and payload weight, and the previously calculated fuel weight.

Table 37. Weights of each high-level component analyzed.

	1st Estimate (lb)	Adjustment (lb)	Weight (lb)
Wing	37,975.9	1,980.2	39,956.2
Empennage	7,014.7	365.8	7,380.4
Fuselage	35,153.9	1,833.1	36,987.0
Nacelles	5,885.9	306.9	6,192.8
Landing Gear	13,868.1	723.1	14,591.2
Power Plant	23,140.3	1,206.6	24,347.0
Fixed Equipment	37,653.4	1,963.4	39,616.8
Empty	160,692.2	8,379.3	169,071.4
Crew			1,230.0
Payload			57,400.0
Fuel			94,811.7
Gross Weight			322,513.1

After determining the weights of each of the components, the longitudinal CG location of each component was defined. The configuration selection of aft-mounted engines was an important factor in determining the CG locations of the wing and nose gear, as it placed significantly more weight towards the rear of the aircraft. The total fuselage length was determined as 189 ft from a previous exercise and was used as the basis for the CG location definitions. The fuselage CG location was set at 87 ft, 46% of the total fuselage length. This approximation was made based on simplified estimations for airliners from Roskam Part V. The CG location for the fixed equipment was assumed to be at the same location as the fuselage. In order to balance the heavy-rear-mounted engines, the components' CG locations are defined more aft. The payload is concentrated slightly behind the fuselage CG, and its CG location is set at 100 ft. The V-tail CG is set as 175 ft and is approximated by using 30% of the mean geometric chord, and to allow enough clearance behind the tail. The engine nacelles are placed slightly in front of the V-tail to avoid interference. The nacelle CG location is set as 40% of the nacelle length and at a location of 165 ft. The wing's CG is set at 115 ft. The root leading edge of the wing is placed at 90.5 ft so that the wing CG location is at 40% of the mean geometric chord (accounting for the wing sweep) of 17.8 ft. The fuel weight CG location is assumed to be approximately similar to the wing CG. Finally, the nose and main landing gear were placed at 15 ft and 120 ft, respectively, to account for the aft-heavy configuration and to meet tip-over requirements. Table 38 summarizes the weight of each component, the component's CG location, and the moment from each component.

Table 38. Longitudinal CG location and moment for each component.

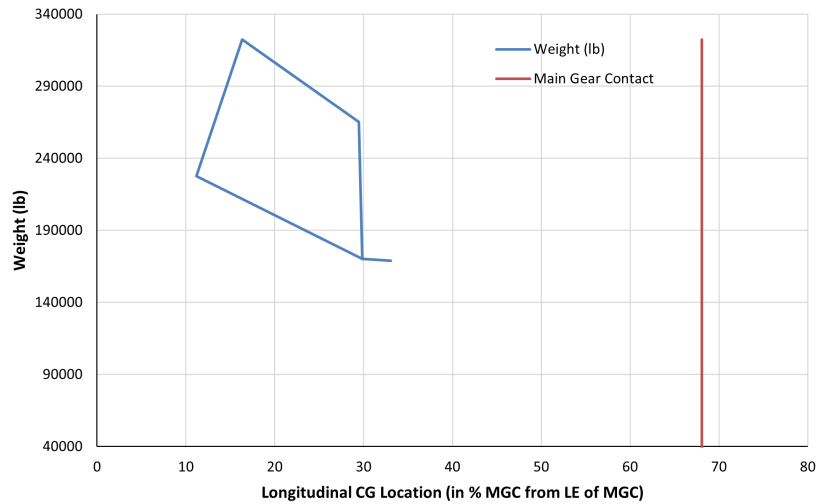
	Longitudinal CG location (ft)	Weight (lb)	Moment (ft-lb)
Wing	115.0	39,956.2	4,594,959.4
Empennage	175.0	7,380.4	1,291,576.8
Fuselage	87.0	36,987.0	3,143,897.2
Nacelles	165.0	6,192.8	1,021,809.0
Nose Gear	15.0	1,021.4	15,320.8
Main Gear	120.0	13,569.8	1,628,379.3
Power Plant	165.0	24,347.0	4,017,249.2
Fixed Equipment	87.0	39,616.8	3,367,431.2
Empty	112.9	169,071.4	
Crew	35.0	1,230.0	30,750.0
Operating Empty	112.2	170,301.4	
Payload	100.0	57,400.0	5,740,000.0
Fuel	113.0	94,811.7	10,713,724.0
Gross Weight	110.3	322,513.1	

8.3 Center of Gravity Envelope

Further using the CG locations of each component, the CG location for the following fuel/payload loadouts is calculated: empty weight (EW), operating empty weight (OEW), operating empty weight + payload (OEW+PLD), total operating gross weight (TOGW), and operating empty weight + fuel (OEW+FUEL). The longitudinal CG locations are then expressed as a percentage of the mean geometric chord (MGC). From here, the CG locations for the main gear and wing were adjusted to values given in Table 38, and to ensure that the CG range is similar to the recommended range of 12%–32% MGC for jet transports. Table 39 shows the resulting CG envelope, which was used to produce the CG excursion diagram shown in Figure 28.

Table 39. Longitudinal axis CG envelope for different fuel/payload loadouts.

	Longitudinal CG location (ft)	Weight (lb)	% MGC
EW	113.76	169,071.4	33.04
OEW	113.19	170,301.4	29.85
OEW+PLD	109.87	227,701.4	11.16
TOGW	110.79	322,513.1	16.34
OEW+FUEL	113.12	265,113.1	29.46

**Figure 28.** Center of gravity excursion diagram.

From Table 39 and Figure 28, it can be seen that the CG travels from 11.16% to 33.04%. Furthermore, the forward-most CG location occurs with operating empty weight + payload, and the aft-most CG location occurs with empty weight. The total operating gross weight CG location is at 16.3% of the mean geometric chord. Figure 29 shows the same CG excursion expressed in terms of absolute fuselage station (X_{cg} in ft). Figure 30 shows a top view of the aircraft and its maximum/minimum CG locations, CG locations of high-level components, and the mean geometric chord line.

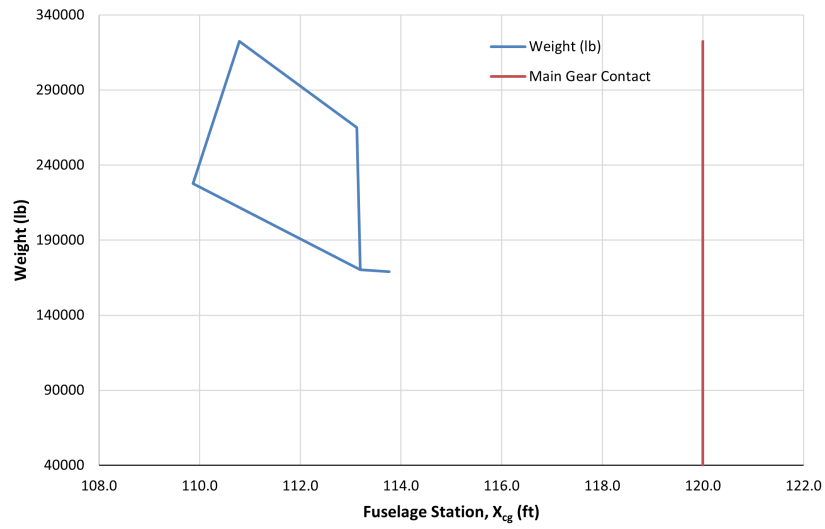


Figure 29. Center of gravity excursion diagram expressed in terms of fuselage station.

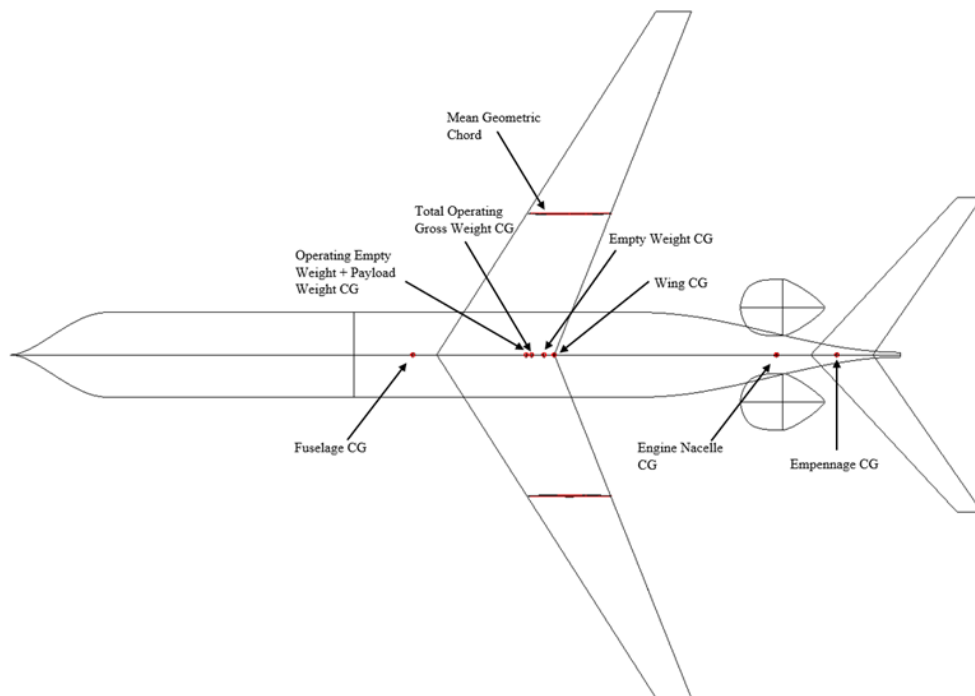


Figure 30. Aircraft top view showing important CG locations.

8.4 Landing Gear Location Analysis

The location of the landing gear is crucial for the plane to prevent longitudinal tip-over and to prevent excess load distribution. The static load on the nose gear should typically fall within a range of 5–15% of the total load, and 85–95% of the total load for the main gear. The load distributions can be calculated at the forward-most and aft-most CG locations of 109.87 ft and 113.76 ft, respectively. The static load on the nose gear can be calculated using Equation (13).

$$\text{Static Load on Nose Gear} = \left(\frac{X_{cg,m} - X_{cg}}{X_{cg,m} - X_{cg,n}} \right) \times 100 \quad (13)$$

Where, $X_{cg,m}$ is the CG of the main gear, X_{cg} is the CG of the aircraft (at specific loadout), and $X_{cg,n}$ is the CG of the nose gear. Table 40 shows the static loads on the nose and main gear for both CG location cases.

Table 40. Static load on nose and main gear at maximum/minimum CG locations.

Longitudinal CG location (ft)	109.87	113.76
Nose Gear Static Load	5.94%	9.65%
Main Gear Static Load	94.06%	90.35%

Table 40 shows that the static load on the landing gear falls within an expected range for the maximum/minimum CG locations. Finally, for the tip-over angle, a vertical center of gravity (in the Z direction) is assumed to be 14 ft. This assumption considers the fuselage diameter of 18 ft (from a previous exercise) and the height increase due to the landing gear, to estimate a CG location close to the centerline of the fuselage. The worst-case tip-over angle can be found using the CG location that is closest to the aft (113.76 ft). Equation (14) is used to find the tip-over angle, where Z_{cg} is the vertical CG.

$$\text{Tip over angle} = \arctan\left(\frac{X_{cg,m} - X_{cg}}{Z_{cg}}\right) \quad (14)$$

The resulting tip-over angle is 24.02°, which exceeds the 15° safety margin for a tricycle gear. Figure 31 shows a side view of the aircraft along with its maximum/minimum CG locations, and the nose and main gear locations.

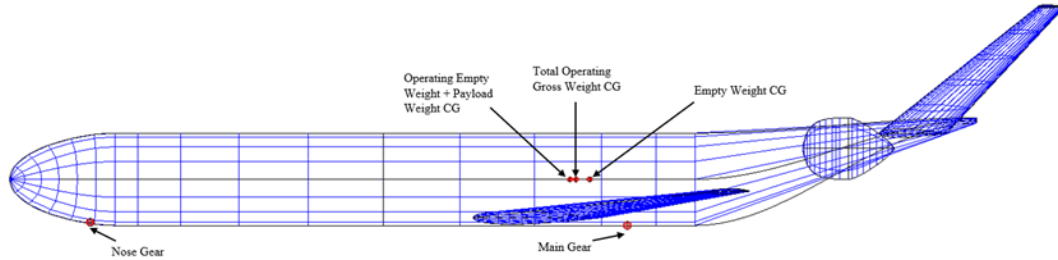


Figure 31. Aircraft side view showing CG locations and nose/main gear locations.

8.5 Weight and Balance Conclusion

The longitudinal weight and balance analysis was performed under the main consideration of the rear-mounted engine, V-tail commercial airliner configuration. The wing and main landing gear were positioned further aft to effectively balance the aircraft. The resulting CG excursion spans between 11.16% and 33.04% mean geometric chord across different fuel/payload loadouts. Roskam Part V recommends an allowable CG range of 12–33% MGC for jet transport aircraft. The forward limit of 11.16% (at OEW + payload) is slightly less than the recommended 12% lower bound, which may warrant further examination of the forward loading case. The aft limit of 33.04% (at empty weight) falls within the recommended upper bound of 33%. These results indicate that the design is broadly within the acceptable range. Finally, the static load distribution maintains a safe load over the nose and main landing gears, and the tip-over angle of 24.02° exceeds the required 15° margin.

9 Conclusion

This report has documented the conceptual design of a middle-of-the-market commercial airliner capable of transporting 280 passengers over a design range of 3,500 nm at a cruise Mach number of 0.80. The design process began with a structured set of mission requirements and progressed through configuration selection, weight and constraint sizing, cost estimation, airfoil and wing analysis, fuselage and tail sizing, and an initial weight and balance analysis. Each stage of the process informed the next, and necessary iterations were made when later analysis resulted in different inputs (most notably the iteration between weight sizing and constraint sizing). The design process resulted in the final state presented below.

9.1 Final Design State

The selected configuration consists of a low-wing, V-tail, twin-turbofan aircraft with engines mounted to the rear fuselage and tricycle landing gear mounted to the wings. This configuration was selected from a weighted Pugh matrix evaluation that prioritized aerodynamic efficiency and passenger ride quality. The final converged design parameters are summarized in Table 41. Figure 32 shows a visualization of the final design state configuration.

Table 41. Final converged design state of the middle-of-the-market commercial airliner.

Parameter	Value
<i>Weights</i>	
Maximum Takeoff Weight (W_{TO})	322,513 lb
Operating Empty Weight (W_{OE})	169,071 lb
Total Fuel Weight (W_f)	94,812 lb
Payload Weight (W_{PL})	57,400 lb
<i>Sizing</i>	
Wing Loading (W_{TO}/S)	136 psf
Thrust-to-Weight Ratio (T_{SL}/W_{TO})	0.28
Total Installed Thrust	90,304 lbf (45,152 lbf per engine)
Cruise L/D	17.36
<i>Wing</i>	
Wing Area (S)	2,371.4 ft ²
Wingspan (b)	146.09 ft
Aspect Ratio (AR)	9
Quarter-Chord Sweep	30°
Taper Ratio	0.3
Root / Tip Chord	24.97 ft / 7.49 ft
Mean Aerodynamic Chord ($\bar{c}$)	17.8 ft
Airfoil	NASA SC(2)-0714
<i>Fuselage</i>	
Fuselage Length (l_f)	189 ft
Fuselage Diameter (d_f)	18 ft
Cabin Layout	8-abreast, single class, 35 rows
<i>Empennage</i>	
V-Tail Configuration	Total area 768.5 ft ²
V-Tail Dihedral	36.5°
V-Tail Aspect Ratio	4.5
V-Tail Airfoil	NACA 0012
<i>Weight and Balance</i>	
CG Excursion Range	11.16% – 33.04% MGC
Tip-Over Angle	24.02°
<i>Cost (at 150 aircraft)</i>	
Estimated Unit Cost	~\$230 million (2026 dollars)

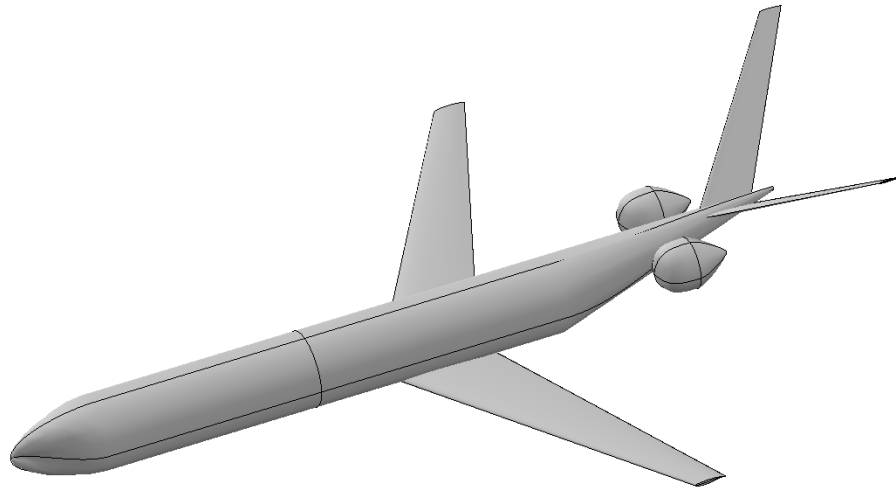


Figure 32. Isometric view of the final commercial airliner configuration

9.2 Limitations and Future Work

The calculations made in this report rely on Class I sizing methods, historical regression data, and simplified aerodynamic models suitable for the conceptual design phase. Limitations and areas for further investigation have been identified throughout the report. The Class I drag polar method does not directly account for changes in wetted area with wing loading, which produced a counterintuitive increase in takeoff weight when the wing loading was increased during the sizing reconvergence. The 3D wing analysis performed in AVL captures induced drag only and does not account for compressibility, viscous, or wave drag effects at the cruise Mach number. Finally, the forward-most CG location of 11.16% MGC is slightly lower than the recommended Roskam range of 12–33% MGC for jet transport aircraft, which warrants further examination of the maximum payload loading case. A more detailed Class II analysis, structural analysis, and stability and control assessment would be required to advance the design from the conceptual to the preliminary design phase.

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